Statistics

Chapter 9 — Instructor Solutions Manual

October 27, 2026 • Total Exercises: 71

Chapter 9 **Exercise 1** Page 305

PROBLEM STATEMENT

Chapter 9, Page 305

Solution A

Step 1

A sample of $n = 49$ observations is taken from a normal population with a standard deviation of $\sigma = 10$. Sample mean is $\bar{X} = 55$.

99 percent Confidence Interval

For a 99 percent level of confidence, the value of z is obtained by locating $0.99/2 = 0.4950$ in the table given in Appendix B1. The value of $z = 2.58$.

$$\begin{aligned} \bar{X} \pm z\frac{\sigma}{\sqrt{n}} &= 55 \pm 2.58 \times \frac{10}{\sqrt{49}} \\ &= 55 \pm 2.58 \times \frac{10}{7} \\ &= 55 \pm 3.69 \end{aligned}$$

So, $X_1 = 55 - 3.69 = 51.31$ and $X_2 = 55 + 3.69 = 58.69$. So, 99 percent confidence interval for population mean is $(51.31, 58.69)$.

Result

$$(51.31, 58.69)$$

PROBLEM STATEMENT

Chapter 9, Page 305

Solution A

Step 1

A sample of $n = 81$ observations is taken from a normal population with a standard deviation of $\sigma = 5$. Sample mean is $\bar{X} = 40$.

95 percent Confidence Interval

For a 95 percent level of confidence, the value of z is obtained by locating $0.95/2 = 0.4750$ in the table given in Appendix B1. The value of $z = 1.96$.

$$\begin{aligned} \bar{X} \pm z\frac{\sigma}{\sqrt{n}} &= 40 \pm 1.96 \times \frac{5}{\sqrt{81}} \\ &= 40 \pm 1.96 \times \frac{5}{9} \\ &= 40 \pm 1.089 \end{aligned}$$

So, $X_1 = 40 - 1.089 = 38.911$ and $X_2 = 40 + 1.089 = 41.089$. So, 95 percent confidence interval for population mean is $(38.911, 41.089)$.

Result

$$(38.911, 41.089)$$

PROBLEM STATEMENT

Chapter 9, Page 305

Solution A

Step 1

A sample of $n = 250$ observations is taken from a normal population with a standard deviation of $\sigma = 25$. Sample mean is $\bar{X} = 20$.

a. Standard error of the mean

$$\begin{aligned}\sigma_{\bar{X}} &= \sigma/\sqrt{n} \\ &= 25/\sqrt{250} \\ &= 25/15.81 \\ &= 1.58\end{aligned}$$

b. The central limit theorem says that the sampling distribution of the mean becomes a normal distribution as sample size increases. A sample of 250 observations is large enough to assume that the sampling distribution will follow the normal distribution.

c. 95 percent Confidence Interval

For a 95 percent level of confidence, the value of z is obtained by locating $0.95/2 = 0.4750$ in the table given in Appendix B1. The value of $z = 1.96$.

$$\begin{aligned}\bar{X} \pm z\frac{\sigma}{\sqrt{n}} &= 20 \pm 1.96 \times \frac{25}{\sqrt{250}} \\ &= 20 \pm 1.96 \times 1.58 \\ &= 20 \pm 3.097\end{aligned}$$

So, $X_1 = 20 - 3.097 = 16.903$ and $X_2 = 20 + 3.097 = 23.097$. So, 95 percent confidence interval for population mean is $(16.903, 23.097)$.

Result

a. 1.58

c. $(16.903, 23.097)$

PROBLEM STATEMENT

Chapter 9, Page 305

Solution A

Step 1

Given:

$$c = \text{Confidence coefficient} = 85\% = 0.85$$

For confidence level $1 - \alpha = 0.85$, determine $z_{\alpha/2} = z_{0.075}$ using the normal probability table in the appendix, which is the z-score closest in value to $0.5000 - 0.075 = 0.4250$.

The closest probability to 0.4250 is 0.4251, which is given in the row starting with "1.4" and in the column starting with ".04", thus the corresponding z-score is $1.4 + .04 = 1.44$:

$$z_{\alpha/2} = 1.44$$

This then implies that the value 1.44 needs to be multiplied by the standard error of the mean.

Result

1.44

PROBLEM STATEMENT

Chapter 9, Page 305

Solution A

Step 1

A sample of $n = 49$ steady smokers is taken from a normal population with a standard deviation of $\sigma = \$5$. Sample mean is $\bar{X} = \$20$.

a. Point estimate of the population mean

The sample mean of $\$20$ is a point estimate of the unknown population mean and is the best estimate, given the data.

b. 95 percent Confidence Interval

For a 95 percent level of confidence, the value of z is obtained by locating $0.95/2 = 0.4750$ in the table given in Appendix B1. The value of $z = 1.96$.

$$\begin{aligned} \bar{X} \pm z\frac{\sigma}{\sqrt{n}} &= 20 \pm 1.96 \times \frac{5}{\sqrt{49}} \\ &= 20 \pm 1.96 \times \frac{5}{7} \\ &= 20 \pm 1.4 \end{aligned}$$

So, $X_1 = 20 - 1.4 = 18.6$ and $X_2 = 20 + 1.4 = 21.4$. So, 95 percent confidence interval for population mean is $(\$18.6, \$21.4)$.

Interpretation: About 95 percent of these confidence intervals are expected to contain the population mean μ i.e mean amount steady smokers spend on cigarettes.

Result

a. $\$20$

b. $(\$18.6, \$21.4)$

PROBLEM STATEMENT

Chapter 9, Page 306

Solution A

Step 1

A sample of $n = 64$ steady smokers is taken from a normal population with a standard deviation of $\sigma = \$5$. Sample mean is $\bar{X} = \$20$.

a. 95 percent Confidence Interval

For a 95 percent level of confidence, the value of z is obtained by locating $0.95/2 = 0.4750$ in the table given in Appendix B1. The value of $z = 1.96$.

$$\begin{aligned} \bar{X} \pm z\frac{\sigma}{\sqrt{n}} &= 20 \pm 1.96 \times \frac{5}{\sqrt{64}} \\ &= 20 \pm 1.96 \times \frac{5}{8} \\ &= 20 \pm 1.225 \end{aligned}$$

So, $X_1 = 20 - 1.225 = 18.775$ and $X_2 = 20 + 1.225 = 21.225$. So, 95 percent confidence interval for population mean is $(\$18.775, \$21.225)$.

b. The new confidence interval is narrower because the sample size has increased. As the sample size increases, the sample mean approaches population mean and as a result, the confidence interval becomes narrower.

Result

a. $(\$18.775, \$21.225)$

PROBLEM STATEMENT

Chapter 9, Page 306

Solution A

Step 1

In finding the point estimate of a given problem, we need to recall the definition of a point estimate. Point estimate is the estimation for the population parameter and it can be represented by the sample mean. Since the population standard deviation is known, we will use the confidence interval formula when σ is known:

$$\overline{x} \pm z\frac{\sigma}{\sqrt{n}}$$

to find the interval from which population mean falls. In addition, we will be using the $z-$table. Based on the obtain interval, we can discuss the estimated mean number of gallons of gasoline sold to Bob's customers.

Step 2

(a) Point Estimate of the Population Mean:

Here, we determine a statistic that will be our estimate for the population parameter. The point estimate will be the sample mean of the data. Since the sample mean of the number of gallons sold to the customers is 8.60 gallons, then the point estimate of the unknown population mean will also be $\boxed{8.60 \text{ gallons}}$.

Step 3

(b) 99% Confidence Interval:

To find the confidence interval with 99% confidence level, we are going to use the formula for the confidence interval for a population mean with σ known:

$$\overline{x} \pm z\frac{\sigma}{\sqrt{n}}$$

Where $\overline{x}$ is the sample mean, z is the $z-$statistic from z table, σ is the standard deviation of the population, and n is the total number of samples.

Now, let us find the values of the quantities involved in the formula. Based on the problem, the mean number of gallons sold is 8.60. Thus,

$$\overline{x} = 8.60$$

The confidence level that we are going to use is 99%. Dividing the confidence level by 2 then locating the obtained value into the $z-$table we have:

$$0.99 \div 2 = 0.4950$$

Then we locate 0.4950 on the $z-$table. 0.4950 or its closest equivalent is found on row 2.5 and column 0.08. The $z-$statistic that corresponds to 0.4951 is will be the sum of the row and column values thus,

$$z = 2.5 + 0.08 = 2.58$$

Step 4

Then, the population standard deviation is known and that is:

$$\sigma = 2.30$$

while the number of samples is

$$n = 60$$

Substituting the values into the confidence interval formula then simplifying we obtain:

$$\begin{aligned}
\overline{x} \pm z\frac{\sigma}{\sqrt{n}} &= 8.60 \pm (2.58)\left(\frac{2.30}{\sqrt{60}}\right) && (\text{Write the formula.}) \\
&= 8.60 \pm (2.58)\left(\frac{2.30}{7.746}\right) && (\text{Evaluate the square root.}) \\
&= 8.60 \pm (2.58)(0.297) && (\text{Divide.}) \\
&= 8.60 \pm 0.766 && (\text{Multiply.})
\end{aligned}$$

Thus, the confidence interval limits are:

$$8.60 - 0.766 = 7.834 \quad \text{and} \quad 8.60 + 0.766 = 9.366$$

Therefore, the 99% confidence interval for the population mean is

$$\boxed{7.834 \text{ to } 9.366}$$

Step 5

(c) Interpretation:

From (b), we calculated the 99% confidence interval for the population mean which is 7.834 to 9.366. This interval indicates that the mean number of gallons of gasoline sold to Bob's customers falls within 7.834 to 9.366 gallons with 99% confidence level. Observe also that the point estimate for the population mean which is 8.60 falls into the interval.

Solution B

Step 1

In finding the point estimate of a given problem, we need to recall the definition of a point estimate. Point estimate is the estimation for the population parameter and it can be represented by the sample mean. Since the population standard deviation is known, we will use the confidence interval formula when σ is known:

$$\overline{x} \pm z\frac{\sigma}{\sqrt{n}}$$

to find the interval from which population mean falls. In addition, we will be using the $z-$table. Based on the obtain interval, we can discuss the estimated mean number of gallons of gasoline sold to Bob's customers.

Step 2

(a) Point Estimate of the Population Mean:

Here, we determine a statistic that will be our estimate for the population parameter. The point estimate will be the sample mean of the data. Since the sample mean of the number of gallons sold to the customers is 8.60 gallons, then the point estimate of the unknown population mean will also be $\boxed{8.60 \text{ gallons}}$.

Step 3

(b) 99% Confidence Interval:

To find the confidence interval with 99% confidence level, we are going to use the formula for the confidence interval for a population mean with σ known:

$$\overline{x} \pm z\frac{\sigma}{\sqrt{n}}$$

Where $\overline{x}$ is the sample mean, z is the $z-$statistic from z table, σ is the standard deviation of the population, and n is the total number of samples.

Now, let us find the values of the quantities involved in the formula. Based on the problem, the mean number of gallons sold is 8.60. Thus,

$$\overline{x} = 8.60$$

The confidence level that we are going to use is 99%. Dividing the confidence level by 2 then locating the obtained value into the $z-$table we have:

$$0.99 \div 2 = 0.4950$$

Then we locate 0.4950 on the $z-$table. 0.4950 or its closest equivalent is found on row 2.5 and column 0.08. The $z-$statistic that corresponds to 0.4951 is will be the sum of the row and column values thus,

$$z = 2.5 + 0.08 = 2.58$$

Step 4

Then, the population standard deviation is known and that is:

$$\sigma = 2.30$$

while the number of samples is

$$n = 60$$

Substituting the values into the confidence interval formula then simplifying we obtain:

$$\begin{aligned}
\overline{x} \pm z\frac{\sigma}{\sqrt{n}} &= 8.60 \pm (2.58)\left(\frac{2.30}{\sqrt{60}}\right) && (\text{Write the formula.}) \\
&= 8.60 \pm (2.58)\left(\frac{2.30}{7.746}\right) && (\text{Evaluate the square root.}) \\
&= 8.60 \pm (2.58)(0.297) && (\text{Divide.}) \\
&= 8.60 \pm 0.766 && (\text{Multiply.})
\end{aligned}$$

Thus, the confidence interval limits are:

$$8.60 - 0.766 = 7.834 \quad \text{and} \quad 8.60 + 0.766 = 9.366$$

Therefore, the 99% confidence interval for the population mean is

$$\boxed{7.834 \text{ to } 9.366}$$

Step 5

(c) Interpretation:

From (b), we calculated the 99% confidence interval for the population mean which is 7.834 to 9.366. This interval indicates that the mean number of gallons of gasoline sold to Bob's customers falls within 7.834 to 9.366 gallons with 99% confidence level. Observe also that the point estimate for the population mean which is 8.60 falls into the interval.

PROBLEM STATEMENT

Chapter 9, Page 306

Solution A

Step 1

A sample of $n = 40$ students is taken from a normal population with a standard deviation of $\sigma = 2.44$ words. Sample mean number of misspelled words is $\bar{X} = 6.05$.

95 percent Confidence Interval

For a 95 percent level of confidence, the value of z is obtained by locating $0.95/2 = 0.4750$ in the table given in Appendix B1. The value of $z = 1.96$.

$$\begin{aligned} \bar{X} \pm z\frac{\sigma}{\sqrt{n}} &= 6.05 \pm 1.96 \times \frac{2.44}{\sqrt{40}} \\ &= 6.05 \pm 1.96 \times \frac{2.44}{6.32} \\ &= 6.05 \pm 0.76 \end{aligned}$$

So, $X_1 = 6.05 - 0.76 = 5.29$ and $X_2 = 6.05 + 0.76 = 6.81$. So, 95 percent confidence interval for population mean is $(5.29, 6.81)$.

Result

(5.29 words, 6.81 words)

PROBLEM STATEMENT

Chapter 9, Page 312

Solution A

Step 1

Recall : The number of degrees of freedom (df) is the number of observations in the sample minus the number of samples, which is written as $n-1$.

$(\mathbf{A.})$ To achieve the goal of the task, first, identify the degrees of freedom with a sample size of 12.

$$\begin{aligned} n &= 12 \\ df &= n-1 \\ &= 12-1 \\ df &= 11 \end{aligned}$$

Step 2

Now, look for the t value on a Student's t Distribution table. Given that $df = 11$ with a confidence interval of 95%.

Upon looking at the Student's t Distribution table the t-value is equal to 2.201.

Step 3

$(\mathbf{B.})$ To achieve the goal of the task, first, identify the degrees of freedom with a sample size of 20.

$$\begin{aligned} n &= 20 \\ df &= n-1 \\ &= 20-1 \\ df &= 19 \end{aligned}$$

Step 4

Now, look for the t value on a Student's t Distribution table. Given that $df = 19$ with a confidence interval of 90%.

Upon looking at the Student's t Distribution table the t-value is equal to 1.729.

Step 5

$(\mathbf{C.})$ To achieve the goal of the task, first, identify the degrees of freedom with a sample size of 8.

$$\begin{aligned} n &= 8 \\ df &= n - 1 \\ &= 8 - 1 \\ df &= 7 \end{aligned}$$

Step 6

Now, look for the t value on a Student's t Distribution table. Given that $df = 7$ with a confidence interval of 99% .

Upon looking at the Student's t Distribution table the t-value is equal to 3.499.

Solution B

Step 1

Recall : The number of degrees of freedom (df) is the number of observations in the sample minus the number of samples, which is written as $n - 1$.

$(\mathbf{A.})$ To achieve the goal of the task, first, identify the degrees of freedom with a sample size of 12.

$$\begin{aligned} n &= 12 \\ df &= n - 1 \\ &= 12 - 1 \\ df &= 11 \end{aligned}$$

Step 2

Now, look for the t value on a Student's t Distribution table. Given that $df = 11$ with a confidence interval of 95%.

Upon looking at the Student's t Distribution table the t-value is equal to 2.201.

Step 3

$(\mathbf{B.})$ To achieve the goal of the task, first, identify the degrees of freedom with a sample size of 20.

$$\begin{aligned} n &= 20 \\ df &= n - 1 \\ &= 20 - 1 \\ df &= 19 \end{aligned}$$

Step 4

Now, look for the t value on a Student's t Distribution table. Given that $df = 19$ with a confidence interval of 90%.

Upon looking at the Student's t Distribution table the t-value is equal to 1.729.

Step 5

$(\mathbf{C.})$ To achieve the goal of the task, first, identify the degrees of freedom with a sample size of 8.

$$\begin{aligned} n &= 8 \\ df &= n - 1 \\ &= 8 - 1 \\ df &= 7 \end{aligned}$$

Step 6

Now, look for the t value on a Student's t Distribution table. Given that $df = 7$ with a confidence interval of 99%

.

Upon looking at the Student's t Distribution table the t-value is equal to 3.499.

Chapter 9 **Exercise 10** Page 312

PROBLEM STATEMENT

Chapter 9, Page 312

Solution A

Step 1

$\textbf{a. Sample size $n = 15$ and level of confidence is 95 percent}$

For a 95 percent confidence interval and degrees of freedom $df = n - 1 = 15 - 1 = 14$, t value is 2.145.

$\textbf{b. Sample size $n = 24$ and level of confidence is 98 percent}$

For a 98 percent confidence interval and degrees of freedom $df = n - 1 = 24 - 1 = 23$, t value is 2.500.

$\textbf{c. Sample size $n = 12$ and level of confidence is 90 percent}$

For a 90 percent confidence interval and degrees of freedom $df = n - 1 = 12 - 1 = 11$, t value is 1.796.

Result

a. 2.145

b. 2.5

c. 1.796

PROBLEM STATEMENT

Chapter 9, Page 312

Solution A

Step 1

A sample of $n = 20$ chickens laid on an average $\bar{X} = 20$ eggs per month with a standard deviation of $s = 2$ eggs per month.

a. Population mean

Population mean is unknown. The best estimate of the population mean is sample mean $\bar{X}$ which is 20 eggs.

b. t distribution is used because population standard deviation is unknown. But to use t distribution, one needs to assume that population follows normal distribution.

$\textbf{c. t value}$

For a 95 percent confidence interval and degrees of freedom $df = n - 1 = 20 - 1 = 19$, t value is 2.093.

Step 2

d. 95 percent Confidence Interval

$$\begin{aligned}\bar{X} \pm t\frac{s}{\sqrt{n}} &= 20 \pm 2.093 \times \frac{2}{\sqrt{20}} \\ &= 20 \pm 2.093 \times 0.45 \\ &= 20 \pm 0.94185\end{aligned}$$

So, $X_1 = 20 - 0.94185 = 19.058$ and $X_2 = 20 + 0.94185 = 20.942$. So, 95 percent confidence interval for population mean is $(19.058, 20.942)$ eggs.

e. 21 does not lie in the confidence interval since it is above the upper limit of the interval. So, population mean is not 21 eggs since it does not lie in the confidence interval.

25 does not lie in the confidence interval since it is above the upper limit of the interval. So, population mean is not 25 eggs since it does not lie in the confidence interval.

🎯 **Result**

a. 20

c. 2.093

d. $(19.058, 20.942)$ eggs

PROBLEM STATEMENT

Chapter 9, Page 313

Solution A

Step 1

A sample of $n = 16$ people reveals the mean yearly consumption to be $\bar{X} = 60$ gallons with a standard deviation of $s = 20$ gallons.

a. Population mean

Population mean is unknown. The best estimate of the population mean is sample mean $\bar{X}$ which is 60 gallons.

b. t distribution is used because population standard deviation is unknown. But to use t distribution, one needs to assume that population follows normal distribution.

$\textbf{c. t value}$

For a 90 percent confidence interval and degrees of freedom $df = n - 1 = 16 - 1 = 15$, t value is 1.753.

Step 2

d. 90 percent Confidence Interval

$$\begin{aligned}\bar{X} \pm t\frac{s}{\sqrt{n}} &= 60 \pm 1.753 \times \frac{20}{\sqrt{16}} \\ &= 60 \pm 1.753 \times 5 \\ &= 60 \pm 8.765\end{aligned}$$

So, $X_1 = 60 - 8.765 = 51.235$ and $X_2 = 60 + 8.765 = 68.765$. So, 95 percent confidence interval for population mean is $(51.235, 68.765)$ gallons.

e. 63 lies in the confidence interval. So, population mean could be 63 gallons since it lies in the confidence interval.

🎯 **Result**

a. 60 gallons

c. 1.753

d. $(51.235, 68.765)$ gallons

PROBLEM STATEMENT

Chapter 9, Page 313

Solution A

Step 1

A sample of $n = 10$ employees who use child care was taken to to estimate the mean weekly child-care cost of their employees.

Population mean estimate

Sample mean $\bar{X}$ is the estimate of population mean.

$$\begin{aligned}\bar{X} =& \frac{1}{10}(107 + 92 + 97 + 95 + 105 + 101 \\ & + 91 + 99 + 95 + 104) \\ =& \frac{986}{10} \\ =& 98.6\end{aligned}$$

Point estimate of the population mean is 98.6

Step 2

Sample variance is

$$\begin{aligned}s^2 =& \frac{\Sigma(X_i - \bar{X})^2}{n} \\ =& \frac{1}{10}\{(107 - 98.6)^2 + (92 - 98.6)^2 + (97 - 98.6)^2 + (95 - 98.6)^2 + (105 - 98.6)^2 \\ & + (101 - 98.6)^2 + (91 - 98.6)^2 + (99 - 98.6)^2 + (95 - 98.6)^2 + (104 - 98.6)^2\} \\ =& \frac{1}{10}\{70.56 + 43.56 + 2.56 + 12.96 + 40.96 + 5.76 + 57.76 + 0.16 + 12.96 + 29.16\} \\ =& \frac{276.4}{10} \\ =& 27.64\end{aligned}$$

Sample standard deviation

$$\begin{aligned}s &= \sqrt{s^2} \\ &= \sqrt{27.64} \\ &= 16.625\end{aligned}$$

Step 3

d. 90 percent Confidence Interval

For a 90 percent confidence interval and degrees of freedom $df = n - 1 = 10 - 1 = 9$, t value is 1.833.

$$\begin{aligned}\bar{X} \pm t\frac{s}{\sqrt{n}} &= 98.6 \pm 1.833 \times \frac{16.625}{\sqrt{10}} \\ &= 98.6 \pm 1.833 \times 5.26 \\ &= 98.6 \pm 9.642\end{aligned}$$

So, $X_1 = 98.6 - 9.642 = 88.958$ and $X_2 = 98.6 + 9.642 = 108.242$. So, 90 percent confidence interval for population mean is $(\$88.958, \$108.242)$.

Interpretation: Out of 100 samples, 90 samples would include the population mean amount spent on the child care.

🎯 Result

$$(\$88.958, \$108.242)$$

PROBLEM STATEMENT

Chapter 9, Page 313

Solution A

Step 1

A sample of $n = 15$ workers was taken to estimate the mean time spent by workers.

Population mean estimate

Sample mean $\bar{X}$ is the estimate of population mean.

$$\begin{aligned}\bar{X} =& \frac{1}{15}(29 + 38 + 38 + 33 + 38 + 21 + 45 \\ & + 34 + 40 + 37 + 37 + 42 + 30 + 29 + 35) \\ =& \frac{526}{15} \\ =& 35.067\end{aligned}$$

Point estimate of the population mean is 35.067

Step 2

Sample variance is

$$\begin{aligned}
s^2 =& \frac{\Sigma(X_i - \bar{X})^2}{n} \\
=& \frac{1}{15}\{(29-35.067)^2 + (38-35.067)^2 + (38-35.067)^2 + (33-35.067)^2 + (38-35.067)^2 \\
& + (21-35.067)^2 + (45-35.067)^2 + (34-35.067)^2 + (40-35.067)^2 + (37-35.067)^2 \\
& + (37-35.067)^2 + (42-35.067)^2 + (30-35.067)^2 + (29-35.067)^2 + (35-35.067)^2\} \\
=& \frac{1}{15}\{36.81 + 8.602 + 8.602 + 4.272 + 8.602 + 197.88 + 98.66 \\
& + 1.138 + 24.334 + 3.736 + 3.736 + 48.066 + 25.674 + 36.81 + 0.0045\} \\
=& \frac{506.9265}{15} \\
=& 33.7951
\end{aligned}$$

Sample standard deviation

$$\begin{aligned}
s &= \sqrt{s^2} \\
&= \sqrt{33.7951} \\
&= 5.813
\end{aligned}$$

Step 3

98 percent Confidence Interval

For a 98 percent confidence interval and degrees of freedom $df = n - 1 = 15 - 1 = 14$, t value is 2.624.

$$\begin{aligned}
\bar{X} \pm t\frac{s}{\sqrt{n}} &= 35.067 \pm 2.624 \times \frac{5.813}{\sqrt{15}} \\
&= 35.067 \pm 2.624 \times 1.501 \\
&= 35.067 \pm 3.94
\end{aligned}$$

So, $X_1 = 35.067 - 3.94 = 31.127$ and $X_2 = 35.067 + 3.94 = 39.007$. So, 98 percent confidence interval for population mean is $(31.127, 39.007)$ minutes.

Interpretation: Out of 100 samples, 98 samples would contain the population mean.

Result

$(31.127, 39.007)$ minutes

Chapter 9 **Exercise 15** Page 316

PROBLEM STATEMENT

Chapter 9, Page 316

Solution A

Step 1

Out of $n = 100$ customers, the owner of the gas station finds that $X = 80$ paid at the pump.

a. Sample proportion is

$$\begin{aligned} p &= \frac{X}{n} \\ &= \frac{80}{100} \\ &= 0.8 \end{aligned}$$

It is estimated that 80% of the population paid at the pump.

b. 95 percent confidence interval

For a 95 percent level of confidence, the value of z is obtained by locating $0.95/2 = 0.4750$ in the table given in Appendix B1. The value of $z = 1.96$.

$$\begin{aligned} p \pm z\sqrt{\frac{p(1-p)}{n}} &= 0.80 \pm 1.96 \times \sqrt{\frac{0.80(1-0.80)}{100}} \\ &= 0.80 \pm 1.96 \times 0.04 \\ &= 0.80 \pm 0.0784 \end{aligned}$$

So, $p_1 = 0.80 - 0.0784 = 0.7216$ and $p_2 = 0.80 + 0.0784 = 0.8784$. So, 95 percent confidence interval for population proportion is $(0.7216, 0.8784)$.

c. Interpretation: 0.8 lies in the confidence interval. So, population proportion lies between 72.16% and 87.84%.

Result

a. 0.8

b. $(0.7216, 0.8784)$

PROBLEM STATEMENT

Chapter 9, Page 316

Solution A

Step 1

Out of $n = 400$ voters, $X = 300$ will support Maria Wilson in the election.

a. Sample proportion is

$$\begin{aligned} p &= \frac{X}{n} \\ &= \frac{300}{400} \\ &= 0.75 \end{aligned}$$

It is estimated that 75% of the population will support Maria Wilson in the election.

b. 99 percent confidence interval

For a 99 percent level of confidence, the value of z is obtained by locating $0.99/2 = 0.4950$ in the table given in Appendix B1. The value of $z = 2.98$.

$$\begin{aligned} p \pm z\sqrt{\frac{p(1-p)}{n}} &= 0.75 \pm 2.98 \times \sqrt{\frac{0.75(1-0.75)}{400}} \\ &= 0.75 \pm 2.98 \times 0.022 \\ &= 0.75 \pm 0.06556 \end{aligned}$$

So, $p_1 = 0.75 - 0.06556 = 0.68444$ and $p_2 = 0.75 + 0.06556 = 0.81556$. So, 99 percent confidence interval for population proportion is $(0.68444, 0.81556)$.

c. Interpretation: The confidence intervals constructed from 99 of the samples would contain the true population proportion.

Result

a. 0.75

b. $(0.68444, 0.81556)$

PROBLEM STATEMENT

Chapter 9, Page 316

Solution A

Step 1

Out of $n = 400$ viewers, $X = 250$ indicated they would watch the new show.

a. Sample proportion is

$$\begin{aligned} p &= \frac{X}{n} \\ &= \frac{250}{400} \\ &= 0.625 \end{aligned}$$

It is estimated that 62.5% of the viewers would watch the new show.

b. 99 percent confidence interval

For a 99 percent level of confidence, the value of z is obtained by locating $0.99/2 = 0.4950$ in the table given in Appendix B1. The value of $z = 2.58$.

$$\begin{aligned} p \pm z\sqrt{\frac{p(1-p)}{n}} &= 0.625 \pm 2.98 \times \sqrt{\frac{0.625(1-0.625)}{400}} \\ &= 0.625 \pm 2.98 \times 0.024 \\ &= 0.625 \pm 0.072 \end{aligned}$$

So, $p_1 = 0.625 - 0.072 = 0.553$ and $p_2 = 0.625 + 0.072 = 0.697$. So, 99 percent confidence interval for population proportion is $(0.553, 0.697)$.

c. Interpretation: The confidence intervals constructed from 99 of the samples would contain the true population proportion.

Result

a. 0.625

b. $(0.553, 0.697)$

PROBLEM STATEMENT

Chapter 9, Page 316

Solution A

Step 1

Out of $n = 300$ cups, $X = 15$ cups were found to be defective.

a. Sample proportion is

$$\begin{aligned} p &= \frac{X}{n} \\ &= \frac{15}{300} \\ &= 0.05 \end{aligned}$$

It is estimated that 5% of the cups were defective.

b. 95 percent confidence interval

For a 95 percent level of confidence, the value of z is obtained by locating $0.95/2 = 0.4750$ in the table given in Appendix B1. The value of $z = 1.96$.

$$\begin{aligned} p \pm z\sqrt{\frac{p(1-p)}{n}} &= 0.05 \pm 1.96 \times \sqrt{\frac{0.05(1-0.05)}{300}} \\ &= 0.05 \pm 1.96 \times 0.0126 \\ &= 0.05 \pm 0.025 \end{aligned}$$

So, $p_1 = 0.05 - 0.025 = 0.025$ and $p_2 = 0.05 + 0.025 = 0.075$. So, 95 percent confidence interval for population proportion is $(0.025, 0.075)$.

c. He should not return the lot since the interval of defective cups is between 2.5% and 7.5% which is below 10%.

Result

a. 0.05

b. $(0.025, 0.075)$

c. He should not return the lot.

PROBLEM STATEMENT

Chapter 9, Page 320

Solution A

Step 1

Given:

$$\sigma = \text{Population standard deviation} = 10$$
$$E = \text{Margin of error} = 2$$
$$c = \text{Confidence coefficient} = 95\% = 0.95$$

Formula sample size:

$$n = \left(\frac{z\sigma}{E}\right)^2$$

For confidence level $1 - \alpha = 0.95$, determine the z-score corresponding to a probability of $1 - 0.5 - \alpha/2 = 1 - 0.5 - 0.05/2 = 0.4750$ using table the normal probability table in the appendix:

$$z = 1.96$$

Evaluate the formula for the sample size (round up to the nearest integer!):

$$n = \left(\frac{z\sigma}{E}\right)^2 = \left(\frac{1.96 \times 10}{2}\right)^2 \approx 97$$

Thus we require a sample of size at least 97.

Result

$$n = 97$$

PROBLEM STATEMENT

Chapter 9, Page 320

Solution A

Step 1

Given:

$$\sigma = \text{Population standard deviation} = 15$$
$$E = \text{Margin of error} = 5$$
$$c = \text{Confidence coefficient} = 99\% = 0.99$$

Formula sample size:

$$n = \left(\frac{z\sigma}{E}\right)^2$$

For confidence level $1 - \alpha = 0.99$, determine the z-score corresponding to a probability of $1 - 0.5 - \alpha/2 = 1 - 0.5 - 0.01/2 = 0.4995$ using table the normal probability table in the appendix:

$$z = 2.58$$

Evaluate the formula for the sample size (round up to the nearest integer!):

$$n = \left(\frac{z\sigma}{E}\right)^2 = \left(\frac{2.58 \times 15}{5}\right)^2 \approx 60$$

Thus we require a sample of size at least 60.

Result

$$n = 60$$

PROBLEM STATEMENT

Chapter 9, Page 320

Solution A

Step 1

Given:

$$\begin{aligned} \pi &= \text{Population proportion} = 0.15 \\ E &= 0.05 \\ c &= \text{Confidence coefficient} = 95\% = 0.95 \end{aligned}$$

Formula sample size:

$$n = \frac{[z_{\alpha/2}]^2 \pi(1-\pi)}{E^2}$$

For confidence level $1-\alpha = 0.95$, determine $z_{\alpha/2} = z_{0.025}$ using the normal probability table in the appendix (look up 0.025 in the table, the z-score is then the found z-score with opposite sign):

$$z_{\alpha/2} = 1.96$$

Let us fill in the known values in the given formula of the sample size and evaluate (round up to the nearest integer!):

$$n = \frac{[z_{\alpha/2}]^2 \pi(1-\pi)}{E^2} = \frac{1.96^2 \times 0.15(1-0.15)}{0.05^2} \approx 196$$

Thus we require a sample of size at least 196.

Result

$$n = 196$$

PROBLEM STATEMENT

Chapter 9, Page 320

Solution A

Step 1

Given:

$$\begin{aligned} \pi &= \text{Population proportion} = 0.45 \\ E &= \text{Margin of error} = 0.10 \\ c &= \text{Confidence coefficient} = 99\% = 0.99 \end{aligned}$$

Formula sample size:

$$n = \frac{z^2 \pi (1-\pi)}{E^2}$$

For confidence level $1-\alpha = 0.99$, determine the z-score corresponding to a probability of $1 - 0.5 - \alpha/2 = 1 - 0.5 - 0.01/2 = 0.4995$ using table the normal probability table in the appendix:

$$z = 2.58$$

Evaluate the formula for the sample size (round up to the nearest integer!):

$$n = \frac{z^2 \pi (1-\pi)}{E^2} = \frac{2.58^2 \times 0.45(1-0.45)}{0.10^2} \approx 165$$

Thus we require a sample of size at least 165.

Result

$$n = 165$$

PROBLEM STATEMENT

Chapter 9, Page 320

Solution A

Step 1

Given:

$$\sigma = \text{Population standard deviation} = 3$$
$$E = \text{Margin of error} = \frac{1}{4} = 0.25$$
$$c = \text{Confidence coefficient} = 95\% = 0.95$$

Formula sample size:

$$n = \left(\frac{z\sigma}{E}\right)^2$$

For confidence level $1 - \alpha = 0.95$, determine the z-score corresponding to a probability of $1 - 0.5 - \alpha/2 = 1 - 0.5 - 0.05/2 = 0.4750$ using table the normal probability table in the appendix:

$$z = 1.96$$

Evaluate the formula for the sample size (round up to the nearest integer!):

$$n = \left(\frac{z\sigma}{E}\right)^2 = \left(\frac{1.96 \times 3}{0.25}\right)^2 \approx 554$$

Thus we should survey at least 554 executives.

Result

554 executives

PROBLEM STATEMENT

Chapter 9, Page 320

Solution A

Step 1

Given:

$$\sigma = \text{Population standard deviation} = 0.5$$
$$E = \text{Margin of error} = 0.2$$
$$c = \text{Confidence coefficient} = 95\% = 0.95$$

Formula sample size:

$$n = \left(\frac{z\sigma}{E}\right)^2$$

For confidence level $1 - \alpha = 0.95$, determine the z-score corresponding to a probability of $1 - 0.5 - \alpha/2 = 1 - 0.5 - 0.05/2 = 0.4750$ using table the normal probability table in the appendix:

$$z = 1.96$$

Evaluate the formula for the sample size (round up to the nearest integer!):

$$n = \left(\frac{z\sigma}{E}\right)^2 = \left(\frac{1.96 \times 0.5}{0.2}\right)^2 \approx 25$$

Thus we require at least 25 boxes.

Result

25 boxes

PROBLEM STATEMENT

Chapter 9, Page 320

Solution A

Step 1

Given:

$$\pi = \text{Population proportion} = 0.60$$
$$E = \text{Margin of error} = 0.04$$
$$c = \text{Confidence coefficient} = 95\% = 0.95$$

(a) Formula sample size:

$$n = \frac{z^2 \pi (1-\pi)}{E^2}$$

For confidence level $1-\alpha = 0.95$, determine the z-score corresponding to a probability of $1 - 0.5 - \alpha/2 = 1 - 0.5 - 0.05/2 = 0.4975$ using table the normal probability table in the appendix:

$$z = 1.96$$

Evaluate the formula for the sample size (round up to the nearest integer!):

$$n = \frac{z^2 \pi (1-\pi)}{E^2} = \frac{1.96^2 \times 0.60(1-0.60)}{0.04^2} \approx 577$$

Thus we require a sample of size at least 577.

Step 2

(b) Formula sample size:

$$n = \frac{z^2 \pi (1 - \pi)}{E^2}$$

For confidence level $1 - \alpha = 0.95$, determine the z-score corresponding to a probability of $1 - 0.5 - \alpha/2 = 1 - 0.5 - 0.05/2 = 0.4975$ using table the normal probability table in the appendix:

$$z = 1.96$$

Since we have no preliminary estimate of the population proportion, we use the population proportion $p = 0.5$ (because this population proportion will result in the largest sample size).

Evaluate the formula for the sample size (round up to the nearest integer!):

$$n = \frac{z^2 \pi (1 - \pi)}{E^2} = \frac{1.96^2 \times 0.5(1 - 0.5)}{0.04^2} \approx 601$$

Thus we require a sample of size at least 601.

Result

(a) $n = 577$

(b) $n = 601$

PROBLEM STATEMENT

Chapter 9, Page 320

Solution A

Step 1

Given:

$$\pi = \text{Population proportion} = 30\% = 0.30$$
$$E = \text{Margin of error} = 1\% = 0.01$$
$$c = \text{Confidence coefficient} = 90\% = 0.90$$

(a) Formula sample size:

$$n = \frac{z^2 \pi (1-\pi)}{E^2}$$

For confidence level $1-\alpha = 0.90$, determine the z-score corresponding to a probability of $1 - 0.5 - \alpha/2 = 1 - 0.5 - 0.10/2 = 0.45$ using table the normal probability table in the appendix:

$$z = 1.645$$

Evaluate the formula for the sample size (round up to the nearest integer!):

$$n = \frac{z^2 \pi (1-\pi)}{E^2} = \frac{1.645^2 \times 0.30(1-0.30)}{0.01^2} \approx 5683$$

Thus we require a sample of size at least 5683.

Step 2

(b) If the sample size decreases, then we contain less information about the population and thus our estimates will be less accurate. If our estimates are less accurate, then our estimate is further from the true value of the population parameter and thus the confidence interval needs to be wider. However, a confidence interval is wider when the margin of error is increased and thus we can decrease the sample size by increasing the margin of error.

It is also possible to decrease the sample size by decreasing the coefficient coefficient.

For example, when we require a margin of error of 0.05 (instead of 0.01), then we obtain a sample size of $n = 228$.

🎯 **Result**

(a) $n = 5683$

(b) Decrease confidence coefficient or increase margin for error.

PROBLEM STATEMENT

Chapter 9, Page 322

Solution A

Step 1

Given:

$$\begin{aligned}\overline{x} &= \text{Sample mean} = 35\\ s &= \text{Sample standard deviation} = 5\\ n &= \text{Sample size} = 36\\ N &= \text{Population size} = 300\\ c &= \text{Confidence level} = 95\% = 0.95\end{aligned}$$

Confidence interval

Determine the t-value by looking in the row starting with degrees of freedom $df = n - 1 = 36 - 1 = 35$ and in the column with $c = 95\%$ in the table of the Student's T distribution:

$$t_{\alpha/2} = 2.030$$

The margin of error including the finite population correction is then:

$$\begin{aligned}E &= t_{\alpha/2} \times \frac{s}{\sqrt{n}}\sqrt{\frac{N-n}{N-1}}\\ &= 2.030 \times \frac{5}{\sqrt{36}}\sqrt{\frac{300-36}{300-1}}\\ &\approx 1.5896\end{aligned}$$

The boundaries of the confidence interval then become:

$$\overline{x} - E = 35 - 1.5896 = 33.4104$$

$$\overline{x} + E = 35 + 1.5896 = 36.5896$$

We are 95% confident that the true population mean is between 33.4104 and 36.5896.

Result

$$(33.4104, 36.5896)$$

PROBLEM STATEMENT

Chapter 9, Page 322

Solution A

Step 1

Given:

$$\begin{aligned}\overline{x} &= \text{Sample mean} = 40\\ s &= \text{Sample standard deviation} = 9\\ n &= \text{Sample size} = 49\\ N &= \text{Population size} = 500\\ c &= \text{Confidence level} = 99\% = 0.99\end{aligned}$$

Confidence interval

Determine the t-value by looking in the row starting with degrees of freedom $df = n - 1 = 49 - 1 = 48$ and in the column with $c = 99\%$ in the table of the Student's T distribution:

$$t_{\alpha/2} = 2.682$$

The margin of error including the finite population correction is then:

$$\begin{aligned}E &= t_{\alpha/2} \times \frac{s}{\sqrt{n}}\sqrt{\frac{N-n}{N-1}}\\ &= 2.682 \times \frac{9}{\sqrt{49}}\sqrt{\frac{500-49}{500-1}}\\ &\approx 3.2782\end{aligned}$$

The boundaries of the confidence interval then become:

$$\overline{x} - E = 40 - 3.2782 = 36.7218$$

$$\overline{x} + E = 40 + 3.2782 = 43.2782$$

We are 99% confident that the true population mean is between 36.7218 and 43.2782.

Result

$$(36.7218, 43.2782)$$

PROBLEM STATEMENT

Chapter 9, Page 322

Solution A

Step 1

Given:

$$\begin{aligned}\overline{x} &= \text{Sample mean} = 1.86\\ s &= \text{Sample standard deviation} = 0.50\\ n &= \text{Sample size} = 50\\ N &= \text{Population size} = 400\\ c &= \text{Confidence level} = 99\% = 0.99\end{aligned}$$

Confidence interval

Determine the t-value by looking in the row starting with degrees of freedom $df = n - 1 = 50 - 1 = 49$ and in the column with $c = 99\%$ in the table of the Student's T distribution:

$$t_{\alpha/2} = 2.680$$

The margin of error including the finite population correction is then:

$$\begin{aligned}E &= t_{\alpha/2} \times \frac{s}{\sqrt{n}}\sqrt{\frac{N-n}{N-1}}\\ &= 2.680 \times \frac{0.50}{\sqrt{50}}\sqrt{\frac{400-50}{400-1}}\\ &\approx 0.1775\end{aligned}$$

The boundaries of the confidence interval then become:

$$\overline{x} - E = 1.86 - 0.1775 = 1.6825$$

$$\overline{x} + E = 1.86 + 0.1775 = 2.0375$$

We are 99% confident that the true population mean is between 1.6825 and 2.0375.

Result

$$(1.6825, 2.0375)$$

PROBLEM STATEMENT

Chapter 9, Page 322

Solution A

Step 1

Given:

$$
\begin{aligned}
x &= \text{Number of successes} = 18 \\
n &= \text{Sample size} = 30 \\
N &= \text{Population size} = 300 \\
c &= \text{Confidence level} = 95\% = 0.95
\end{aligned}
$$

Confidence interval

The sample proportion is the number of successes divided by the sample size:

$$\pi = \frac{x}{n} = \frac{18}{30} \approx 0.6$$

For confidence level $1 - \alpha = 0.95$, determine the z-score corresponding to a probability of $1 - 0.5 - \alpha/2 = 1 - 0.5 - 0.05/2 = 0.4975$ using table the normal probability table in the appendix:

$$z = 1.96$$

The margin of error including the finite population correction is then:

$$
\begin{aligned}
E &= z \times \sqrt{\frac{\pi(1-\pi)}{n}} \sqrt{\frac{N-n}{N-1}} \\
&= 1.96 \times \sqrt{\frac{0.6(1-0.6)}{30}} \sqrt{\frac{300-30}{300-1}} \\
&\approx 0.1666
\end{aligned}
$$

The boundaries of the confidence interval then become:

$$\pi - E = 0.6 - 0.1666 = 0.4334$$

$$\pi + E = 0.6 + 0.1666 = 0.7666$$

We are 95% confident that the true population proportion is between 0.4334 and 0.7666.

Result

$$(0.4334, 0.7666)$$

Chapter 9 **Exercise 31** Page 323

PROBLEM STATEMENT

Chapter 9, Page 323

Solution A

Step 1

A sample of $n = 85$ leaders is taken from a normal population with a standard deviation of $\sigma = 1.7$ years. Sample mean time spent on the job is $\bar{X} = 6.5$ years.

95 percent Confidence Interval

For a 95 percent level of confidence, the value of z is obtained by locating $0.95/2 = 0.4750$ in the table given in Appendix B1. The value of $z = 1.96$.

$$\begin{aligned} \bar{X} \pm z\frac{\sigma}{\sqrt{n}} &= 6.5 \pm 1.96 \times \frac{1.7}{\sqrt{85}} \\ &= 6.5 \pm 1.96 \times \frac{1.7}{9.22} \\ &= 6.5 \pm 0.361 \end{aligned}$$

So, $X_1 = 6.5 - 0.361 = 6.139$ and $X_2 = 6.5 + 0.361 = 6.861$. So, 95 percent confidence interval for mean time spent on the job is $(6.139, 6.861)$ hours.

Result

(6.139, 6.861) hours

PROBLEM STATEMENT

Chapter 9, Page 324

Solution A

Step 1

A sample of $n = 36$ packages is taken from a normal population with a standard deviation of $\sigma = 0.03$ pounds. Sample mean weight is $\bar{X} = 3.01$ pounds.

a. Population mean estimate

Estimated population mean is the sample mean which is 3.01 pounds.

b. 95 percent Confidence Interval

For a 95 percent level of confidence, the value of z is obtained by locating $0.95/2 = 0.4750$ in the table given in Appendix B1. The value of $z = 1.96$.

$$\begin{aligned} \bar{X} \pm z\frac{\sigma}{\sqrt{n}} &= 3.01 \pm 1.96 \times \frac{0.03}{\sqrt{36}} \\ &= 3.01 \pm 1.96 \times \frac{0.03}{6} \\ &= 3.01 \pm 0.0098 \end{aligned}$$

So, $X_1 = 3.01 - 0.0098 = 3.0002$ and $X_2 = 3.01 + 0.0098 = 3.0198$. So, 95 percent confidence interval for population mean is $(3.0002, 3.0198)$ pounds.

Result

a. 3.01 pounds

b. (3.002 pounds, 3.0198 pounds)

PROBLEM STATEMENT

Chapter 9, Page 324

Solution A

Step 1

A sample of $n = 40$ showed the mean cost per month was $\bar{X} = \$323$ with a standard deviation of $s = \$25$.

a. 98 percent Confidence Interval

For a 98 percent confidence interval and degrees of freedom $df = n - 1 = 40 - 1 = 39$, t value is 2.426.

$$\begin{aligned}\bar{X} \pm t\frac{s}{\sqrt{n}} &= 323 \pm 2.426 \times \frac{25}{\sqrt{40}} \\ &= 323 \pm 2.426 \times 3.953 \\ &= 323 \pm 9.59\end{aligned}$$

So, $X_1 = 323 - 9.59 = 313.41$ and $X_2 = 323 + 9.59 = 332.59$. So, 98 percent confidence interval for population mean is $(\$313.41, \$332.59)$.

b. 350 does not lie in the confidence interval since it is above the upper limit of the interval. So, it is unreasonable to conclude that population mean is $\$350$ per month since it does not lie in the confidence interval.

Result

a. $(\$313.41, \$332.59)$

PROBLEM STATEMENT

Chapter 9, Page 324

Solution A

Step 1

A sample of $n = 50$ laid off executives it took a mean of $\bar{X} = 26$ weeks to find another position, with a standard deviation of $s = 6.2$ weeks.

95 percent Confidence Interval

For a 95 percent confidence interval and degrees of freedom $df = n - 1 = 50 - 1 = 49$, t value is 2.010.

$$\begin{aligned} \bar{X} \pm t\frac{s}{\sqrt{n}} &= 26 \pm 2.010 \times \frac{6.2}{\sqrt{50}} \\ &= 26 \pm 2.093 \times 0.88 \\ &= 26 \pm 1.842 \end{aligned}$$

So, $X_1 = 26 - 1.842 = 24.158$ and $X_2 = 26 + 1.842 = 27.842$. So, 95 percent confidence interval for population mean is $(24.158, 27.842)$ weeks.

28 does not lie in the confidence interval since it is above the upper limit of the interval. So, it is unreasonable to conclude that population mean is 28 weeks since it does not lie in the confidence interval.

Result

$(24.158, 27.842)$ weeks

PROBLEM STATEMENT

Chapter 9, Page 324

Solution A

Step 1

A sample of $n = 40$ members of YWCA have a mean length of membership of $\bar{X} = 8.32$ years with a standard deviation of $s = 3.07$ years.

a. Population mean

Population mean is unknown. The best estimate of the population mean is sample mean $\bar{X}$ which is 8.32 years.

b. 90 percent Confidence Interval

For a 90 percent confidence interval and degrees of freedom $df = n - 1 = 40 - 1 = 39$, t value is 1.685.

$$\begin{aligned}\bar{X} \pm t\frac{s}{\sqrt{n}} &= 8.32 \pm 1.685 \times \frac{3.07}{\sqrt{40}} \\ &= 8.32 \pm 1.685 \times 0.486 \\ &= 8.32 \pm 0.819\end{aligned}$$

So, $X_1 = 8.32 - 0.819 = 7.501$ and $X_2 = 8.32 + 0.819 = 9.139$. So, 90 percent confidence interval for population mean is $(7.501, 9.139)$ years.

c. 10 does not lie in the confidence interval since it is above the upper limit of the interval. So, it is unreasonable to say that the population mean length of membership is almost 10 years since it does not lie in the confidence interval.

Result

a. Sample mean 8.32 years

b. $(7.501, 9.139)$ years

PROBLEM STATEMENT

Chapter 9, Page 324

Solution A

Step 1

Given:

$$\begin{aligned}\overline{x} &= \text{Sample mean} = 2.76\\ s &= \text{Sample standard deviation} = 0.75\\ n &= \text{Sample size} = 60\\ c &= \text{Confidence level} = 97\% = 0.97\end{aligned}$$

Confidence interval

Determine the t-value by looking in the row starting with degrees of freedom $df = n - 1 = 60 - 1 = 59$ and in the column with $c = 97\%$ in the table of the Student's T distribution. \color{#4257b2}Note: The table in the back of the book does not contain a column with $c = 97\%$ and thus I had to use a *Student t-distribution calculator* to calculate the t-value.

$$t_{\alpha/2} = 2.224$$

Step 2

The margin of error is then:

$$\begin{aligned}E &= t_{\alpha/2} \times \frac{s}{\sqrt{n}}\\ &= 2.224 \times \frac{0.75}{\sqrt{60}}\\ &\approx 0.2153\end{aligned}$$

The boundaries of the confidence interval then become:

$$\overline{x} - E = 2.76 - 1.5896 = 1.1704$$

$$\overline{x} + E = 2.76 + 1.5896 = 4.3496$$

We are 97% confident that the true population mean of number of meals eaten outside the home is between 1.1704 meals and 4.3496 meals.

Result

$$(1.1704, 4.3496)$$

Chapter 9 Exercise 37

Page 324

PROBLEM STATEMENT

Chapter 9, Page 324

Solution A

Step 1

A sample of $n = 50$ assistant showed that the sample mean number of hours spent per week on coaching $\bar{X} = 68.6$ with a standard deviation of $s = 8.2$ hours per week. Population mean is estimated to be $\mu = 70$ hours.

a. 99 percent Confidence Interval

For a 99 percent confidence interval and degrees of freedom $df = n - 1 = 50 - 1 = 49$, t value is 2.680.

$$\begin{aligned} \bar{X} \pm t\frac{s}{\sqrt{n}} &= 68.6 \pm 2.680 \times \frac{8.2}{\sqrt{50}} \\ &= 68.6 \pm 2.680 \times 1.16 \\ &= 68.6 \pm 3.11 \end{aligned}$$

So, $X_1 = 68.6 - 3.11 = 65.49$ and $X_2 = 68.6 + 3.11 = 71.71$. So, 99 percent confidence interval for population mean is $(65.49, 71.71)$ hours.

b. 70 lies in the 99 percent confidence interval $(65.49, 71.71)$. So, population mean is not 21 eggs since it does not lie in the confidence interval. So, population mean could be 70 hours per week.

c. Changing the level of confidence from 99 percent to 95 percent would increase the width of the confidence interval. The change in the value of t from 2.680 to 2.010 would increase the width of the interval.

Result

a. $(65.49, 71.71)$ hours

c. Increase the interval

Chapter 9 **Exercise 38** Page 324

PROBLEM STATEMENT

Chapter 9, Page 324

Solution A

Step 1

A sample of $n = 45$ employees reveals the mean amount spent on dental expenses was $\bar{X} = \$1820$ with a standard deviation of $s = \$660$.

a. 95 percent Confidence Interval

For a 95 percent confidence interval and degrees of freedom $df = n - 1 = 45 - 1 = 44$, t value is 2.015.

$$\begin{aligned} \bar{X} \pm t\frac{s}{\sqrt{n}} &= 1820 \pm 2.015 \times \frac{660}{\sqrt{45}} \\ &= 1820 \pm 2.015 \times 98.36 \\ &= 1820 \pm 198.19 \end{aligned}$$

So, $X_1 = 1820 - 198.19 = 1621.81$ and $X_2 = 1820 + 198.19 = 2018.19$. So, 95 percent confidence interval for population mean is $(\$1621.81, \$2018.19)$.

b. 1700 lies in the 95 percent confidence interval $(\$1621.81, \$2018.19)$. So, it is possible that the population mean could be $\$1700$.

Result

a. $(\$1621.81, \$2018.19)$

PROBLEM STATEMENT

Chapter 9, Page 324

Solution A

Step 1

Given:

$$\begin{aligned} s &= \text{Sample standard deviation} = 50 \\ E &= \text{Margin of error} = 2 \\ c &= \text{Confidence coefficient} = 95\% = 0.95 \end{aligned}$$

Formula sample size:

$$n = \left(\frac{z\sigma}{E}\right)^2$$

In this case, the population standard deviation is unknown, thus we should replace σ by s (which is appropriate as the sample size is at least 30).

$$n = \left(\frac{zs}{E}\right)^2$$

For confidence level $1 - \alpha = 0.95$, determine the z-score corresponding to a probability of $1 - 0.5 - \alpha/2 = 1 - 0.5 - 0.05/2 = 0.4750$ using table the normal probability table in the appendix:

$$z = 1.96$$

Evaluate the formula for the sample size:

$$n = \left(\frac{zs}{E}\right)^2 = \left(\frac{1.96 \times 50}{2}\right)^2 = 2401$$

Thus the sample size was 2401.

Result

$$n = 2401$$

PROBLEM STATEMENT

Chapter 9, Page 324

Solution A

Step 1

A sample of $n = 20$ dealers revealed that the mean amount of profit per car sold was $\bar{X} = \$290$ with a standard deviation of $s = \$125$.

95 percent Confidence Interval

For a 95 percent confidence interval and degrees of freedom $df = n - 1 = 20 - 1 = 19$, t value is 2.093.

$$\begin{aligned} \bar{X} \pm t\frac{s}{\sqrt{n}} &= 290 \pm 2.093 \times \frac{125}{\sqrt{20}} \\ &= 290 \pm 2.093 \times 27.96 \\ &= 290 \pm 58.52 \end{aligned}$$

So, $X_1 = 290 - 58.52 = 231.48$ and $X_2 = 290 + 58.52 = 348.52$. So, 95 percent confidence interval for population mean is $(\$231.48, \$348.52)$.

Result

$$(\$231.48, \$348.52)$$

PROBLEM STATEMENT

Chapter 9, Page 324

Solution A

Step 1

A sample of $n = 25$ graduates revealed that the mean amount owed by a student in student loans was $\bar{X} = \$14,381$ with a standard deviation of $s = \$1892$ eggs per month.

90 percent Confidence Interval

For a 90 percent confidence interval and degrees of freedom $df = n - 1 = 25 - 1 = 24$, t value is 1.711.

$$\begin{aligned}\bar{X} \pm t\frac{s}{\sqrt{n}} &= 14381 \pm 1.711 \times \frac{1892}{\sqrt{25}} \\ &= 14381 \pm 1.711 \times \frac{1892}{5} \\ &= 14381 \pm 647.4424\end{aligned}$$

So, $X_1 = 14381 - 647.4424 = 13733.56$ \ and $X_2 = 14381 + 647.4424 = 15028.44$.

So, 90 percent confidence interval for population mean is $(\$13733.56, \$15028.44)$.

15000 lies in the 90 percent confidence interval $(13733.56, 15028.44)$. So, it is reasonable to assume that the population mean is $\$15000$.

Result

$$(\$13733.56, \$15028.44)$$

PROBLEM STATEMENT

Chapter 9, Page 325

Solution A

Step 1

A sample of $n = 15$ homes sold revealed that the mean number looking through each home was $\bar{X} = 24$ with a standard deviation of $s = 5$ people.

For a 98 percent confidence interval and degrees of freedom $df = n - 1 = 15 - 1 = 14$, t value is 2.624.

98 percent Confidence Interval

$$\begin{aligned} \bar{X} \pm t\frac{s}{\sqrt{n}} &= 24 \pm 2.624 \times \frac{5}{\sqrt{15}} \\ &= 24 \pm 2.624 \times 1.292 \\ &= 24 \pm 3.39 \end{aligned}$$

So, $X_1 = 24 - 3.39 = 20.618$ and $X_2 = 24 + 3.39 = 27.39$. So, 98 percent confidence interval for population mean is $(20.618, 27.39)$.

Result

$$(20.618, 27.39)$$

PROBLEM STATEMENT

Chapter 9, Page 325

Solution A

Step 1

A sample of $n = 12$ subscribers was taken to check if typical customer spends $\mu = \$60$ per month on local and long-distance service.

a. Population mean estimate

Sample mean $\bar{X}$ is the estimate of population mean.

$$\begin{aligned}\bar{X} =& \frac{1}{12}(64 + 66 + 64 + 66 + 59 + 62 + 67 \\ & + 61 + 64 + 58 + 54 + 66) \\ =& \frac{751}{12} \\ =& 62.583\end{aligned}$$

Point estimate of the population mean is 62.583

Step 2

Sample variance is

$$\begin{aligned}
s^2 =& \frac{\Sigma(X_i - \bar{X})^2}{n} \\
=& \frac{1}{12}\{(64 - 62.583)^2 + (66 - 62.583)^2 + (64 - 62.583)^2 + (64 - 62.583)^2 + (66 - 62.583)^2 \\
& + (62 - 62.583)^2 + (67 - 62.583)^2 + (61 - 62.583)^2 + (64 - 62.583)^2 + (58 - 62.583)^2 \\
& + (54 - 62.583)^2 + (66 - 62.583)^2\} \\
=& \frac{1}{12}\{2.008 + 11.676 + 2.008 + 11.676 + 12.838 + 0.34 + 19.51 \\
& + 2.506 + 2.008 + 21.004 + 73.67 + 11.66 \\
=& \frac{170.904}{12} \\
=& 14.242
\end{aligned}$$

Sample standard deviation is

$$\begin{aligned}
s &= \sqrt{s^2} \\
&= \sqrt{14.242} \\
&= 3.774
\end{aligned}$$

Step 3

b. 90 percent Confidence Interval

For a 90 percent confidence interval and degrees of freedom $df = n - 1 = 12 - 1 = 11$, t value is 1.796.

$$\begin{aligned}
\bar{X} \pm t\frac{s}{\sqrt{n}} &= 62.583 \pm 1.796 \times \frac{3.774}{\sqrt{12}} \\
&= 62.583 \pm 1.796 \times 1.089 \\
&= 62.583 \pm 1.956
\end{aligned}$$

So, $X_1 = 62.583 - 1.956 = 60.627$ and $X_2 = 62.583 + 1.956 = 64.539$. So, 90 percent confidence interval for population mean is $(\$60.627, \$64.539)$.

c. 60 does not lie in the confidence interval since it is below the lower limit of the interval. So, it is unreasonable to claim that the “typical customer” spends $\$60$ per month.

🎯 Result

a, $\$62.583$

b.$(\$60.627, \$64.539)$

c. No

PROBLEM STATEMENT

Chapter 9, Page 325

Solution A

Step 1

A sample of $n = 10$ catridges was taken to to advertise the number of pages a user can expect from a print cartridge.

a. Population mean estimate

Sample mean $\bar{X}$ is the estimate of population mean.

$$\begin{aligned}\bar{X} =& \frac{1}{10}(2698 + 2028 + 2474 + 2395 + 2372 + 2475 \\ &+ 1927 + 3006 + 2334 + 2379) \\ =& \frac{24088}{10} \\ =& 2408.8\end{aligned}$$

Point estimate of the population mean is 2408.8

Step 2

Sample variance is

$$\begin{aligned}
s^2 =& \frac{\Sigma(X_i - \bar{X})^2}{n} \\
=& \frac{1}{10}\{(2698 - 2408.8)^2 + (2028 - 2408.8)^2 + (2474 - 2408.8)^2 + (2395 - 2408.8)^2 \\
&+ (2372 - 2408.8)^2 + (2475 - 2408.8)^2 + (1927 - 2408.8)^2 + (3006 - 2408.8)^2 \\
&+ (2334 - 2408.8)^2 + (2379 - 2408.8)^2\} \\
=& \frac{1}{10}\{83636.64 + 145008.64 + 4251.04 + 190.44 + 1354.24 \\
&+ 4382.44 + 232131.24 + 3566647.84 + 5595.04 + 888.04\} \\
=& \frac{4044085.6}{10} \\
=& 404408.56
\end{aligned}$$

Sample standard deviation is

$$\begin{aligned}
s &= \sqrt{s^2} \\
&= \sqrt{404408.56} \\
&= 635.93
\end{aligned}$$

Step 3

b. 95 percent Confidence Interval

For a 95 percent confidence interval and degrees of freedom $df = n - 1 = 10 - 1 = 9$, t value is 2.262.

$$\begin{aligned}
\bar{X} \pm t\frac{s}{\sqrt{n}} &= 2408.8 \pm 2.262 \times \frac{635.93}{\sqrt{10}} \\
&= 2408.8 \pm 2.262 \times 201.12 \\
&= 2408.8 \pm 454.93
\end{aligned}$$

So, $X_1 = 2408.8 - 454.93 = 1953.87$ and $X_2 = 2048.8 + 454.93 = 2863.73$. So, 95 percent confidence interval for population mean is $(1953.87, 2863.73)$.

Result

a. 2408.8

b. $(1953.87, 2863.73)$

PROBLEM STATEMENT

Chapter 9, Page 325

Solution A

Step 1

A sample of $n = 15$ executives to check the stress level scores.

a. Population mean estimate

Sample mean $\bar{X}$ is the estimate of population mean.

$$\begin{aligned}\bar{X} =& \frac{1}{15}(94 + 78 + 83 + 90 + 78 + 99 + 97 \\ & + 90 + 97 + 90 + 93 + 94 + 100 + 75 + 84) \\ =& \frac{1312}{15} \\ =& 87.467\end{aligned}$$

Point estimate of the population mean is 87.467

Step 2

Sample variance is

$$\begin{aligned}
s^2 =& \frac{\Sigma(X_i - \bar{X})^2}{n} \\
=& \frac{1}{15}\{(94 - 87.467)^2 + (78 - 87.467)^2 + (83 - 87.467)^2 + (90 - 87.467)^2 + (78 - 87.467)^2 \\
& + (99 - 87.467)^2 + (97 - 87.467)^2 + (90 - 87.467)^2 + (97 - 87.467)^2 + (90 - 87.467)^2 \\
& + (93 - 87.467)^2 + (94 - 87.467)^2 + (100 - 87.467)^2 + (75 - 87.467)^2 + (84 - 87.467)^2\} \\
=& \frac{1}{15}\{42.68 + 89.624 + 19.954 + 6.416 + 89.624 + 133.01 + 90.878 \\
& + 6.416 + 90.878 + 6.416 + 30.614 + 42.68 + 157.076 + 155.426 + 12.02\} \\
=& \frac{973.712}{15} \\
=& 64.91413
\end{aligned}$$

Sample standard deviation is

$$\begin{aligned}
s &= \sqrt{s^2} \\
&= \sqrt{64.91413} \\
&= 8.057
\end{aligned}$$

Step 3

b. 95 percent Confidence Interval

For a 95 percent confidence interval and degrees of freedom $df = n - 1 = 15 - 1 = 14$, t value is 2.145.

$$\begin{aligned}
\bar{X} \pm t\frac{s}{\sqrt{n}} &= 87.467 \pm 2.145 \times \frac{8.057}{\sqrt{15}} \\
&= 87.467 \pm 2.145 \times 0.45 \\
&= 87.467 \pm 8.3076
\end{aligned}$$

So, $X_1 = 87.467 - 8.3076 = 79.1594$ and $X_2 = 87.467 + 8.3076 = 95.7746$. So, 95 percent confidence interval for population mean is $(79.1594, 95.7746)$.

Step 4

c. Dangerous Stress level score 80 lies in the 95 percent confidence interval $(79.1594, 95.7746)$. So, it reasonable to conclude that Internet executives have a mean stress level in the dangerous level.

🎯 **Result**

a. 87.467

b. $(79.1594, 95.7746)$

c. Yes

PROBLEM STATEMENT

Chapter 9, Page 325

Solution A

Step 1

Out of $n = 220$ applicants, $X = 14$ failed the drug test.

Sample proportion is

$$\begin{aligned} p &= \frac{X}{n} \\ &= \frac{14}{220} \\ &= 0.0636 \end{aligned}$$

It is estimated that 6.36% of the applicants failed the drug taste.

99 percent confidence interval

For a 99 percent level of confidence, the value of z is obtained by locating $0.99/2 = 0.4950$ in the table given in Appendix B1. The value of $z = 2.58$.

$$\begin{aligned} p \pm z\sqrt{\frac{p(1-p)}{n}} &= 0.0636 \pm 2.58 \times \sqrt{\frac{0.0636(1-0.0636)}{220}} \\ &= 0.0636 \pm 2.58 \times 0.0164 \\ &= 0.0636 \pm 0.0423 \end{aligned}$$

So, $p_1 = 0.0636 - 0.0423 = 0.0213$ and $p_2 = 0.0636 + 0.0423 = 0.1059$. So, 99 percent confidence interval for population proportion is $(0.0213, 0.1059)$.

The percentage of applicants failing the drug test is between 2.13% to 10.59%. which is more than 10%. So, it is incorrect to conclude that more than 10% of the applicants are now failing the test.

Result

$$(0.0213, 0.1059)$$

PROBLEM STATEMENT

Chapter 9, Page 325

Solution A

Step 1

Out of $n = 400$ tests, $X = 14$ employees failed the test.

Sample proportion is

$$\begin{aligned} p &= \frac{X}{n} \\ &= \frac{14}{400} \\ &= 0.035 \end{aligned}$$

It is estimated that 3.5% of the employees failed the taste.

99 percent confidence interval

For a 99 percent level of confidence, the value of z is obtained by locating $0.99/2 = 0.4950$ in the table given in Appendix B1. The value of $z = 2.58$.

$$\begin{aligned} p \pm z\sqrt{\frac{p(1-p)}{n}} &= 0.035 \pm 2.58 \times \sqrt{\frac{0.035(1-0.035)}{400}} \\ &= 0.035 \pm 2.58 \times 0.0092 \\ &= 0.035 \pm 0.024 \end{aligned}$$

So, $p_1 = 0.035 - 0.024 = 0.011$ and $p_2 = 0.035 + 0.024 = 0.059$. So, 99 percent confidence interval for population proportion is $(0.011, 0.059)$.

The percentage of employees failing the test is between 1.11% to 5.59%. which implies 5% is included in the interval. So, it would not be reasonable to conclude that less than 5% of the employees are now failing the test.

Result

$$(0.011, 0.059)$$

PROBLEM STATEMENT

Chapter 9, Page 325

Solution A

Step 1

Out of $n = 500$ small business owners, $p = 65\%$ or %0.65$of small business owners do not approve the health care changes.\

\textbf{95 percent confidence interval }\ For a 95 percent level of confidence, the value of$z$is obtained by locating $0.95/2 = 0.4750$in the table given in Appendix B1. The value of$z=1.96

$$p \pm z\sqrt{\frac{p(1-p)}{n}} = 0.65 \pm 1.96 \times \sqrt{\frac{0.65(1-0.65)}{500}} \quad So,$$
$$= 0.65 \pm 1.96 \times 0.0213$$
$$= 0.65 \pm 0.042$$

.

p_1 =0.65-0.042=0.608andp_2=0.65+0.042=0.692$. So,$95$percent confidence interval for population proportion is$(0.608, 0.692)$.

Interpretation: Out of 100 different samples, 95 samples would contain the proportion of owners opposing the health changes.

Result

$$(0.608, 0.692)$$

PROBLEM STATEMENT

Chapter 9, Page 325

Solution A

Step 1

Out of $n = 500$ voters, $X = 350$ plan to vote to return Miller.

Sample proportion is

$$\begin{aligned} p &= \frac{X}{n} \\ &= \frac{350}{500} \\ &= 0.7 \end{aligned}$$

It is estimated that 70% of the population will vote for Miller.

99 percent confidence interval

For a 99 percent level of confidence, the value of z is obtained by locating $0.99/2 = 0.4950$ in the table given in Appendix B1. The value of $z = 2.58$.

$$\begin{aligned} p \pm z\sqrt{\frac{p(1-p)}{n}} &= 0.7 \pm 2.58 \times \sqrt{\frac{0.7(1-0.7)}{500}} \\ &= 0.7 \pm 2.58 \times 0.0205 \\ &= 0.7 \pm 0.05289 \end{aligned}$$

So, $p_1 = 0.7 - 0.05289 = 0.64711$ and $p_2 = 0.7 + 0.05289 = 0.75289$. So, 99 percent confidence interval for population proportion is $(0.64711, 0.75289)$.

Result

$$(0.6471, 0.75289)$$

PROBLEM STATEMENT

Chapter 9, Page 325

Solution A

Step 1

Given:

$$x = \text{Number of successes} = 560$$
$$n = \text{Sample size} = 1000$$
$$c = \text{Confidence level} = 95\% = 0.95$$

(a) **Confidence interval**

The sample proportion is the number of successes divided by the sample size:

$$\pi = \frac{x}{n} = \frac{560}{1000} = 0.560$$

For confidence level $1 - \alpha = 0.95$, determine the z-score corresponding to a probability of $1 - 0.5 - \alpha/2 = 1 - 0.5 - 0.05/2 = 0.4975$ using table the normal probability table in the appendix:

$$z = 1.96$$

The margin of error is then:

$$\begin{aligned} E &= z \times \sqrt{\frac{\pi(1-\pi)}{n}} \\ &= 1.96 \times \sqrt{\frac{0.560(1-0.560)}{1000}} \\ &\approx 0.0308 \end{aligned}$$

The boundaries of the confidence interval then become:

$$\pi - E = 0.560 - 0.0308 = 0.5292$$

$$\pi + E = 0.560 + 0.0308 = 0.5908$$

We are 95% confident that the true population proportion of respondents who feel the president is doing a good job is between 0.5292 and 0.5908.

Step 2

(b) Confidence interval found in part (a): $(0.5292, 0.5908)$

The majority implies more than 50% or equivalently more than 0.50.

Since the confidence interval contains only values above 0.50, it is reasonable to conclude that a majority of the population believes the president is doing a good job.

🎯 Result

(a) $(0.5292, 0.5908)$

(b) Yes

PROBLEM STATEMENT

Chapter 9, Page 326

Solution A

Step 1

Given:

$$\begin{aligned}\overline{x} &= \text{Sample mean} = 54\\ s &= \text{Sample standard deviation} = 4.50\\ n &= \text{Sample size} = 35\\ N &= \text{Population size} = 500\\ c &= \text{Confidence level} = 95\% = 0.95\end{aligned}$$

Confidence interval

Determine the t-value by looking in the row starting with degrees of freedom $df = n - 1 = 35 - 1 = 34$ and in the column with $c = 95\%$ in the table of the Student's T distribution:

$$t_{\alpha/2} = 2.032$$

The margin of error including the finite population correction is then:

$$\begin{aligned}E &= t_{\alpha/2} \times \frac{s}{\sqrt{n}}\sqrt{\frac{N-n}{N-1}}\\ &= 2.032 \times \frac{4.50}{\sqrt{35}}\sqrt{\frac{500-35}{500-1}}\\ &\approx 1.4920\end{aligned}$$

The boundaries of the confidence interval then become:

$$\overline{x} - E = 54 - 1.4920 = 52.5080$$

$$\overline{x} + E = 54 + 1.4920 = 55.4920$$

We are 95% confident that the true population mean amount of a citation in River City is between $\$52.5080$ and $\$55.4920$.

🎯 Result

$$(52.5080, 55.4920)$$

PROBLEM STATEMENT

Chapter 9, Page 326

Solution A

Step 1

Given:

$$\begin{aligned} x &= \text{Number of successes} = 26 \\ n &= \text{Sample size} = 50 \\ N &= \text{Population size} = 650 \\ c &= \text{Confidence level} = 99\% = 0.99 \end{aligned}$$

Confidence interval

The sample proportion is the number of successes divided by the sample size:

$$\pi = \frac{x}{n} = \frac{26}{50} \approx 0.52$$

For confidence level $1 - \alpha = 0.99$, determine the z-score corresponding to a probability of $1 - 0.5 - \alpha/2 = 1 - 0.5 - 0.01/2 = 0.4995$ using table the normal probability table in the appendix:

$$z = 2.58$$

The margin of error including the finite population correction is then:

$$\begin{aligned} E &= z \times \sqrt{\frac{\pi(1-\pi)}{n}} \\ &= 2.58 \times \sqrt{\frac{0.52(1-0.52)}{50}} \\ &\approx 0.182 \end{aligned}$$

The boundaries of the confidence interval then become:

$$\pi - E = 0.52 - 0.182 = 0.338$$

$$\pi + E = 0.52 + 0.182 = 0.702$$

We are 99% confident that the true population proportion of checking account customers who have a Visa card with the bank is between 0.338 and 0.702.

Result

$$(0.338, 0.702)$$

Chapter 9 **Exercise 53** Page 326

PROBLEM STATEMENT

Chapter 9, Page 326

Solution A

Step 1

Given:

$$\pi = \text{Population proportion} = 60\% = 0.60$$
$$E = \text{Margin of error} = 5\% = 0.05$$
$$c = \text{Confidence coefficient} = 95\% = 0.95$$

Formula sample size:

$$n = \frac{z^2 \pi (1 - \pi)}{E^2}$$

For confidence level $1 - \alpha = 0.95$, determine the z-score corresponding to a probability of $1 - 0.5 - \alpha/2 = 1 - 0.5 - 0.05/2 = 0.4975$ using table the normal probability table in the appendix:

$$z = 1.96$$

Evaluate the formula for the sample size (round up to the nearest integer!):

$$n = \frac{z^2 \pi (1 - \pi)}{E^2} = \frac{1.96^2 \times 0.60(1 - 0.60)}{0.05^2} \approx 369$$

Thus we require a sample of size at least 369.

Result

$$n = 369$$

Chapter 9 **Exercise 54** Page 326

PROBLEM STATEMENT

Chapter 9, Page 326

Solution A

Step 1

Given:

$$\begin{aligned}\sigma &= \text{Population standard deviation} = 14\\ E &= \text{Margin of error} = 2\\ c &= \text{Confidence coefficient} = 90\% = 0.90\end{aligned}$$

Formula sample size:

$$n = \left(\frac{z\sigma}{E}\right)^2$$

For confidence level $1-\alpha = 0.90$, determine the z-score corresponding to a probability of $1-0.5-\alpha/2 = 1-0.5-0.10/2 = 0.4500$ using table the normal probability table in the appendix:

$$z = 1.645$$

Evaluate the formula for the sample size (round up to the nearest integer!):

$$n = \left(\frac{z\sigma}{E}\right)^2 = \left(\frac{1.645 \times 14}{2}\right)^2 \approx 133$$

Thus we require a sample of size at least 133.

Result

133 outside salespeople

PROBLEM STATEMENT

Chapter 9, Page 326

Solution A

Step 1

Given:

$$\begin{aligned} \sigma &= \text{Population standard deviation} \approx 500 \\ E &= \text{Margin of error} = 100 \\ c &= \text{Confidence coefficient} = 95\% = 0.95 \end{aligned}$$

Formula sample size:

$$n = \left(\frac{z\sigma}{E}\right)^2$$

For confidence level $1 - \alpha = 0.95$, determine the z-score corresponding to a probability of $1 - 0.5 - \alpha/2 = 1 - 0.5 - 0.05/2 = 0.4750$ using table the normal probability table in the appendix:

$$z = 1.96$$

Evaluate the formula for the sample size (round up to the nearest integer!):

$$n = \left(\frac{z\sigma}{E}\right)^2 = \left(\frac{1.96 \times 500}{100}\right)^2 \approx 97$$

Thus we require a sample of size at least 97 and thus at least 97 families need to be interviewed.

Result

97 families

PROBLEM STATEMENT

Chapter 9, Page 326

Solution A

Step 1

(a) Given:

$$\begin{aligned}\overline{x} &= \text{Sample mean} = 10979\\ s &= \text{Sample standard deviation} = 1000\\ n &= \text{Sample size} = 20\\ c &= \text{Confidence level} = 90\% = 0.90\end{aligned}$$

Confidence interval

Determine the t-value by looking in the row starting with degrees of freedom $df = n - 1 = 20 - 1 = 19$ and in the column with $c = 90\%$ in the table of the Student's T distribution:

$$t_{\alpha/2} = 1.729$$

The margin of error is then:

$$\begin{aligned}E &= t_{\alpha/2} \times \frac{s}{\sqrt{n}}\\ &= 1.729 \times \frac{1000}{\sqrt{20}}\\ &\approx 386.6162\end{aligned}$$

The boundaries of the confidence interval then become:

$$\overline{x} - E = 10979 - 386.6162 = 10592.3838$$

$$\overline{x} + E = 10979 + 386.6162 = 11365.6162$$

We are 90% confident that the true population mean yearly premium is between $\$10,592.3838$ and $\$11,365.6162$.

Step 2

(b) Given:

$$\begin{aligned}\sigma &= \text{Population standard deviation} = 1000\\ E &= \text{Margin of error} = 250\\ c &= \text{Confidence coefficient} = 99\% = 0.99\end{aligned}$$

Formula sample size:

$$n = \left(\frac{z\sigma}{E}\right)^2$$

For confidence level $1-\alpha = 0.99$, determine the z-score corresponding to a probability of $1-0.5-\alpha/2 = 1-0.5-0.01/2 = 0.4950$ using table the normal probability table in the appendix:

$$z = 2.58$$

Evaluate the formula for the sample size (round up to the nearest integer!):

$$n = \left(\frac{z\sigma}{E}\right)^2 = \left(\frac{2.58 \times 1000}{250}\right)^2 \approx 107$$

Thus we require a sample of size at least 107.

Result

(a) $(10592.3838, 11365.6162)$

(b) $n = 107$

PROBLEM STATEMENT

Chapter 9, Page 326

Solution A

Step 1

(a) Given:

$$\begin{aligned}\overline{x} &= \text{Sample mean} = 8000\\ s &= \text{Sample standard deviation} = 300\\ n &= \text{Sample size} = 30\\ c &= \text{Confidence level} = 99\% = 0.99\end{aligned}$$

Confidence interval

Determine the t-value by looking in the row starting with degrees of freedom $df = n - 1 = 30 - 1 = 29$ and in the column with $c = 99\%$ in the table of the Student's T distribution:

$$t_{\alpha/2} = 2.756$$

The margin of error is then:

$$\begin{aligned}E &= t_{\alpha/2} \times \frac{s}{\sqrt{n}}\\ &= 2.756 \times \frac{300}{\sqrt{30}}\\ &\approx 150.9523\end{aligned}$$

The boundaries of the confidence interval then become:

$$\overline{x} - E = 8000 - 150.9523 = 7849.0477$$

$$\overline{x} + E = 8000 + 150.9523 = 8150.9523$$

We are 90% confident that the true population mean equivalent pressure is between 7849.0477 feet and 8150.9523 feet.

Step 2

(b) Given:

$$\begin{aligned}\sigma &= \text{Population standard deviation} = 300\\ E &= \text{Margin of error} = 25\\ c &= \text{Confidence coefficient} = 95\% = 0.95\end{aligned}$$

Formula sample size:

$$n = \left(\frac{z\sigma}{E}\right)^2$$

For confidence level $1-\alpha = 0.95$, determine the z-score corresponding to a probability of $1-0.5-\alpha/2 = 1-0.5-0.05/2 = 0.4750$ using table the normal probability table in the appendix:

$$z = 1.96$$

Evaluate the formula for the sample size (round up to the nearest integer!):

$$n = \left(\frac{z\sigma}{E}\right)^2 = \left(\frac{1.96\times 300}{25}\right)^2 \approx 554$$

Thus we require a sample of size at least 554.

Result

(a) $(7849.0477, 8150.9523)$

(b) $n = 554$

PROBLEM STATEMENT

Chapter 9, Page 326

Solution A

Step 1

(a) The population mean is the mean wage of all employees in the Florida state.

The best estimate of the population mean is the sample mean, which has been given as $\$65.00$ per hour.

Step 2

(b) Given:

$$\begin{aligned}\overline{x} &= \text{Sample mean} = 65.00\\ s &= \text{Sample standard deviation} = 6.25\\ n &= \text{Sample size} = 25\\ c &= \text{Confidence level} = 99\% = 0.99\end{aligned}$$

Confidence interval

Determine the t-value by looking in the row starting with degrees of freedom $df = n - 1 = 25 - 1 = 24$ and in the column with $c = 99\%$ in the table of the Student's T distribution:

$$t_{\alpha/2} = 2.797$$

The margin of error is then:

$$\begin{aligned}E &= t_{\alpha/2} \times \frac{s}{\sqrt{n}}\\ &= 2.797 \times \frac{6.25}{\sqrt{25}}\\ &\approx 3.49625\end{aligned}$$

The boundaries of the confidence interval then become:

$$\overline{x} - E = 65.00 - 3.49625 = 61.50375$$

$$\overline{x} + E = 65.00 + 3.49625 = 68.49625$$

We are 99% confident that the true population mean wage for these employees is between $\$61.50375$ and $\$68.49625$.

Step 3

(c) Given:

$$\begin{aligned}\sigma &= \text{Population standard deviation} = 6.25\\ E &= \text{Margin of error} = 1.00\\ c &= \text{Confidence coefficient} = 95\% = 0.95\end{aligned}$$

Formula sample size:

$$n = \left(\frac{z\sigma}{E}\right)^2$$

For confidence level $1-\alpha = 0.95$, determine the z-score corresponding to a probability of $1-0.5-\alpha/2 = 1-0.5-0.05/2 = 0.4750$ using table the normal probability table in the appendix:

$$z = 1.96$$

Evaluate the formula for the sample size (round up to the nearest integer!):

$$n = \left(\frac{z\sigma}{E}\right)^2 = \left(\frac{1.96 \times 6.25}{1.00}\right)^2 \approx 151$$

Thus we require a sample of size at least 151.

Result

(a) $\$65.00$ per hour

(b) $(61.50375, 68.49625)$

(c) $n = 151$

PROBLEM STATEMENT

Chapter 9, Page 326

Solution A

Step 1

(a) Given:

$$\begin{aligned}\overline{x} &= \text{Sample mean} = 78\\ s &= \text{Sample standard deviation} = 9\\ n &= \text{Sample size} = 50\\ c &= \text{Confidence level} = 95\% = 0.95\end{aligned}$$

Confidence interval

Determine the t-value by looking in the row starting with degrees of freedom $df = n - 1 = 50 - 1 = 49$ and in the column with $c = 95\%$ in the table of the Student's T distribution:

$$t_{\alpha/2} = 2.010$$

The margin of error is then:

$$\begin{aligned}E &= t_{\alpha/2} \times \frac{s}{\sqrt{n}}\\ &= 2.010 \times \frac{9}{\sqrt{50}}\\ &\approx 2.5583\end{aligned}$$

The boundaries of the confidence interval then become:

$$\overline{x} - E = 78 - 2.5583 = 75.4417$$

$$\overline{x} + E = 78 + 2.5583 = 80.5583$$

We are 95% confident that the true population mean number of hours spent watching videos and DVDs last year is between 75.4417 hours and 80.5583 hours.

Step 2

(b) Given:

$$\begin{aligned}\sigma &= \text{Population standard deviation} \approx 9\\ E &= \text{Margin of error} = 1.0\\ c &= \text{Confidence coefficient} = 90\% = 0.90\end{aligned}$$

Formula sample size:

$$n = \left(\frac{z\sigma}{E}\right)^2$$

For confidence level $1-\alpha = 0.90$, determine the z-score corresponding to a probability of $1-0.5-\alpha/2 = 1-0.5-0.10/2 = 0.4500$ using table the normal probability table in the appendix:

$$z = 1.65$$

Evaluate the formula for the sample size (round up to the nearest integer!):

$$n = \left(\frac{z\sigma}{E}\right)^2 = \left(\frac{1.65 \times 9}{1.0}\right)^2 \approx 221$$

Thus we require a sample of size at least 221.

Result

(a) $(75.4417, 80.5583)$

(b) $n = 221$

PROBLEM STATEMENT

Chapter 9, Page 326

Solution A

Step 1

(a) Given:

$$\begin{aligned}\overline{x} &= \text{Sample mean} = 23.4\\ s &= \text{Sample standard deviation} = 0.9\\ n &= \text{Sample size} = 9\\ c &= \text{Confidence level} = 95\% = 0.95\end{aligned}$$

Confidence interval

Determine the t-value by looking in the row starting with degrees of freedom $df = n - 1 = 9 - 1 = 8$ and in the column with $c = 95\%$ in the table of the Student's T distribution:

$$t_{\alpha/2} = 2.306$$

The margin of error is then:

$$\begin{aligned}E &= t_{\alpha/2} \times \frac{s}{\sqrt{n}}\\ &= 2.306 \times \frac{0.9}{\sqrt{9}}\\ &= 0.6918\end{aligned}$$

The boundaries of the confidence interval then become:

$$\overline{x} - E = 23.4 - 0.6918 = 22.7082$$

$$\overline{x} + E = 23.4 + 0.6918 = 24.0918$$

We are 90% confident that the true population mean mpg is between 22.7082 mpg and 24.0918 mpg.

Step 2

(b) Given:

$$\begin{aligned}\sigma &= \text{Population standard deviation} = 0.9\\ E &= \text{Margin of error} = 0.1\\ c &= \text{Confidence coefficient} = 95\% = 0.95\end{aligned}$$

Formula sample size:

$$n = \left(\frac{z\sigma}{E}\right)^2$$

For confidence level $1-\alpha = 0.95$, determine the z-score corresponding to a probability of $1-0.5-\alpha/2 = 1-0.5-0.01/2 = 0.4750$ using table the normal probability table in the appendix:

$$z = 1.96$$

Evaluate the formula for the sample size (round up to the nearest integer!):

$$n = \left(\frac{z\sigma}{E}\right)^2 = \left(\frac{1.96\times 0.9}{0.1}\right)^2 \approx 312$$

Thus we require a sample of size at least 312 and thus he should fill up his gas tank 312 times.

Result

(a) $(22.7082, 24.0918)$

(b) 312 times

PROBLEM STATEMENT

Chapter 9, Page 327

Solution A

Step 1

(a) Given:

$$s = \text{Sample standard deviation} = 180$$
$$n = \text{Sample size} = 36$$

The standard error of the sample mean is the sample standard deviation s divided by the square root of the sample size n.

$$SE = \frac{s}{\sqrt{n}} = \frac{180}{\sqrt{36}} = \frac{180}{6} = 30$$

Thus the standard error of the sample mean is $\$30$.

Step 2

(b) Given:

$$\begin{aligned}\overline{x} &= \text{Sample mean} = 416\\ s &= \text{Sample standard deviation} = 180\\ n &= \text{Sample size} = 36\\ c &= \text{Confidence level} = 95\% = 0.95\end{aligned}$$

Confidence interval

Determine the t-value by looking in the row starting with degrees of freedom $df = n - 1 = 36 - 1 = 35$ and in the column with $c = 95\%$ in the table of the Student's T distribution:

$$t_{\alpha/2} = 2.030$$

The margin of error is then:

$$\begin{aligned}E &= t_{\alpha/2} \times \frac{s}{\sqrt{n}}\\ &= 2.030 \times \frac{180}{\sqrt{36}}\\ &= 60.9\end{aligned}$$

The boundaries of the confidence interval then become:

$$\overline{x} - E = 416 - 60.9 = 355.1$$

$$\overline{x} + E = 416 + 60.9 = 476.9$$

We are 95% confident that the true population mean purchase price is between $\$355.10$ and $\$476.90$.

Step 3

(c) Given:

$$\begin{aligned}\sigma &= \text{Population standard deviation} \approx 180\\ E &= \text{Margin of error} = 10\\ c &= \text{Confidence coefficient} = 95\% = 0.95 \quad \text{(assumption)}\end{aligned}$$

Formula sample size:

$$n = \left(\frac{z\sigma}{E}\right)^2$$

For confidence level $1-\alpha = 0.95$, determine the z-score corresponding to a probability of $1-0.5-\alpha/2 = 1-0.5-0.05/2 = 0.4750$ using table the normal probability table in the appendix:

$$z = 1.96$$

Evaluate the formula for the sample size (round up to the nearest integer!):

$$n = \left(\frac{z\sigma}{E}\right)^2 = \left(\frac{1.96 \times 180}{10}\right)^2 \approx 1245$$

Thus we require a sample of size at least 1245.

Result

(a) $\$30$

(b) $(\$355.10, \$476.90)$

(c) $n = 1245$

PROBLEM STATEMENT

Chapter 9, Page 327

Solution A

Step 1

Given:

$$\pi = \text{Population proportion} = \frac{5}{50} = \frac{1}{10} = 0.1$$
$$E = \text{Margin of error} = 2\% = 0.02$$
$$c = \text{Confidence coefficient} = 95\% = 0.95$$

Formula sample size:

$$n = \frac{z^2\pi(1-\pi)}{E^2}$$

For confidence level $1-\alpha = 0.95$, determine the z-score corresponding to a probability of $1 - 0.5 - \alpha/2 = 1 - 0.5 - 0.05/2 = 0.4750$ using table the normal probability table in the appendix:

$$z = 1.96$$

Evaluate the formula for the sample size (round up to the nearest integer!):

$$n = \frac{z^2\pi(1-\pi)}{E^2} = \frac{1.96^2 \times 0.1(1-0.1)}{0.02^2} \approx 865$$

Thus we require a sample of size at least 865.

Result

$$n = 865$$

PROBLEM STATEMENT

Chapter 9, Page 327

Solution A

Step 1

(a) Given:

$$\begin{aligned} \pi &= \text{Population proportion} = 21\% = 0.21 \\ E &= \text{Margin of error} = 3\% = 0.03 \\ c &= \text{Confidence coefficient} = 95\% = 0.95 \end{aligned}$$

Formula sample size:

$$n = \frac{z^2 \pi (1-\pi)}{E^2}$$

For confidence level $1-\alpha = 0.95$, determine the z-score corresponding to a probability of $1 - 0.5 - \alpha/2 = 1 - 0.5 - 0.05/2 = 0.4750$ using table the normal probability table in the appendix:

$$z = 1.96$$

Evaluate the formula for the sample size (round up to the nearest integer!):

$$n = \frac{z^2 \pi (1-\pi)}{E^2} = \frac{1.96^2 \times 0.21(1-0.21)}{0.03^2} \approx 709$$

Thus we require a sample of size at least 709.

Step 2

(b) Given:

$$E = \text{Margin of error} = 3\% = 0.03$$
$$c = \text{Confidence coefficient} = 95\% = 0.95$$

Formula sample size:

$$n = \frac{z^2 \pi (1 - \pi)}{E^2}$$

For confidence level $1 - \alpha = 0.95$, determine the z-score corresponding to a probability of $1 - 0.5 - \alpha/2 = 1 - 0.5 - 0.05/2 = 0.4750$ using table the normal probability table in the appendix:

$$z = 1.96$$

Since we have no preliminary estimate of the population proportion, we use the population proportion $\pi = 50\% = 0.50$ as this proportion will lead to the largest possible sample size (which is thus safe to use for all population proportions).

Evaluate the formula for the sample size (round up to the nearest integer!):

$$n = \frac{z^2 \pi (1 - \pi)}{E^2} = \frac{1.96^2 \times 0.50(1 - 0.50)}{0.03^2} \approx 1068$$

Thus we require a sample of size at least 1068.

Result

(a) $n = 709$

(b) $n = 1068$

PROBLEM STATEMENT

Chapter 9, Page 327

Solution A

Step 1

Given:

$$\begin{aligned}\overline{x} &= \text{Sample mean} = 32000\\ s &= \text{Sample standard deviation} = 8200\\ n &= \text{Sample size} = 36\\ c &= \text{Confidence level} = 90\% = 0.90\end{aligned}$$

Confidence interval

Determine the t-value by looking in the row starting with degrees of freedom $df = n - 1 = 36 - 1 = 35$ and in the column with $c = 90\%$ in the table of the Student's T distribution:

$$t_{\alpha/2} = 1.690$$

The margin of error is then:

$$\begin{aligned}E &= t_{\alpha/2} \times \frac{s}{\sqrt{n}}\\ &= 1.690 \times \frac{8200}{\sqrt{36}}\\ &\approx 2309.6667\end{aligned}$$

The boundaries of the confidence interval then become:

$$\overline{x} - E = 32000 - 2309.6667 = 29690.3333$$

$$\overline{x} + E = 32000 + 2309.6667 = 34309.6667$$

We are 90% confident that the true population mean account valuation of the population of customers is between $\$29,690.3333$ and $\$34,309.6667$.

Result

$$(\$29690.3333, \$34309.6667)$$

PROBLEM STATEMENT

Chapter 9, Page 327

Solution A

Step 1

(a) Given:

$$x = \text{Number of successes} = 459$$
$$n = \text{Sample size} = 2700$$
$$c = \text{Confidence level} = 95\% = 0.95$$

Confidence interval

The sample proportion is the number of successes divided by the sample size:

$$\pi = \frac{x}{n} = \frac{459}{2700} \approx 0.17$$

For confidence level $1 - \alpha = 0.95$, determine the z-score corresponding to a probability of $1 - 0.5 - \alpha/2 = 1 - 0.5 - 0.05/2 = 0.4750$ using table the normal probability table in the appendix:

$$z = 1.96$$

The margin of error is then:

$$\begin{aligned} E &= z \times \sqrt{\frac{\pi(1-\pi)}{n}} \\ &= 1.96 \times \sqrt{\frac{0.17(1-0.17)}{2700}} \\ &\approx 0.0142 \end{aligned}$$

The boundaries of the confidence interval then become:

$$\pi - E = 0.17 - 0.0142 = 0.1558$$

$$\pi + E = 0.17 + 0.0142 = 0.1842$$

We are 95% confident that the true population proportion of people on low-carbohydrate diets is between 0.1558 and 0.1842.

Step 2

(b) Result part (a): $(0.1558, 0.1842)$

We note that the value $18\% = 0.18$ lies within the confidence interval, which implies that it is likely that the population proportion is 0.18 and thus it is likely that the population percentage is 18%.

Thus it is possible that the population percentage is 18%.

Step 3

(c) Given:

$$\pi = \text{Population proportion} = \frac{459}{2700} \approx 0.17$$
$$E = \text{Margin of error} = 0.5\% = 0.005$$
$$c = \text{Confidence coefficient} = 95\% = 0.95$$

Formula sample size:

$$n = \frac{z^2 \pi (1 - \pi)}{E^2}$$

For confidence level $1 - \alpha = 0.95$, determine the z-score corresponding to a probability of $1 - 0.5 - \alpha/2 = 1 - 0.5 - 0.05/2 = 0.4750$ using table the normal probability table in the appendix:

$$z = 1.96$$

Evaluate the formula for the sample size (round up to the nearest integer!):

$$n = \frac{z^2 \pi (1 - \pi)}{E^2} = \frac{1.96^2 \times 0.17(1 - 0.17)}{0.005^2} \approx 21,682$$

Thus we require a sample of size at least 21,682.

Result

(a) $(0.1558, 0.1842)$

(b) Yes

(c) $n = 21,682$

PROBLEM STATEMENT

Chapter 9, Page 327

Solution A

Step 1

Given:

$$\pi = \text{Sample proportion} = 52\% = 0.52$$
$$n = \text{Sample size} = 1000$$
$$c = \text{Confidence level} = 95\% = 0.95$$

(a) **Confidence interval**

For confidence level $1 - \alpha = 0.95$, determine the z-score corresponding to a probability of $1 - 0.5 - \alpha/2 = 1 - 0.5 - 0.05/2 = 0.4975$ using table the normal probability table in the appendix:

$$z = 1.96$$

The margin of error is then:

$$\begin{aligned} E &= z \times \sqrt{\frac{\pi(1-\pi)}{n}} \\ &= 1.96 \times \sqrt{\frac{0.52(1-0.52)}{1000}} \\ &\approx 0.0310 \end{aligned}$$

The boundaries of the confidence interval then become:

$$\pi - E = 0.52 - 0.0310 = 0.4890$$

$$\pi + E = 0.52 + 0.0310 = 0.5510$$

We are 95% confident that the true population proportion favoring the Republican candidate is between 0.4890 and 0.5510.

Step 2

(b) We assume that the population proportion is the 52% mentioned in the exercise prompt (else we won't be able to estimate the probability).

$$p = \text{Population proportion} = 52\% = 0.52$$

The mean of the sampling distribution of the sample proportions $\hat{p}$ is equal to the population proportion p.

$$\mu_\pi = p = 52\% = 0.52$$

The standard deviation of the sampling distribution of the sample proportion $\hat{p}$ is the square root of the product of p and $(1-p)$ divided by the sample size n.

$$\sigma_\pi = \sqrt{\frac{p(1-p)}{n}} = \sqrt{\frac{0.52(1-0.52)}{1000}} \approx 0.0158$$

The Democratic candidate is leading when the proportion favoring the Republican candidate is actually less than $50\% = 0.05$.

The z-score is the value decreased by the mean, divided by the standard deviation.

$$z = \frac{x-\mu}{\sigma} = \frac{0.50-0.52}{0.0158} \approx -1.27$$

Determine the corresponding probability using the normal probability table in the appendix. $P(0 < Z < 1.27)$ is given in the row starting with 1.2 and in the column starting with .07 of the standard normal probability table in the appendix.

$$\begin{aligned} P(\hat{p} < 0.50) &= P(Z < -1.27) \\ &= P(Z > 1.27) \\ &= 1 - P(Z < 1.27) \\ &= 1 - P(Z < 0) - P(0 < Z < 1.27) \\ &= 1 - 0.5000 - 0.3980 \\ &= 0.1020 \\ &= 10.20\% \end{aligned}$$

Step 3

(c) Given:

$$\begin{aligned}\pi &= \text{Sample proportion} = 52\% = 0.52\\ n &= \text{Sample size} = 3000\\ c &= \text{Confidence level} = 95\% = 0.95\end{aligned}$$

Confidence interval

For confidence level $1-\alpha = 0.95$, determine the z-score corresponding to a probability of $1-0.5-\alpha/2 = 1-0.5-0.05/2 = 0.4975$ using table the normal probability table in the appendix:

$$z = 1.96$$

The margin of error is then:

$$\begin{aligned}E &= z \times \sqrt{\frac{\pi(1-\pi)}{n}}\\ &= 1.96 \times \sqrt{\frac{0.52(1-0.52)}{3000}}\\ &\approx 0.0179\end{aligned}$$

The boundaries of the confidence interval then become:

$$\pi - E = 0.52 - 0.0179 = 0.5021$$

$$\pi + E = 0.52 + 0.0179 = 0.5379$$

We are 95% confident that the true population proportion favoring the Republican candidate is between 0.5021 and 0.5379.

Step 4

Probability

We assume that the population proportion is the 52% mentioned in the exercise prompt (else we won't be able to estimate the probability).

$$p = \text{Population proportion} = 52\% = 0.52$$

The mean of the sampling distribution of the sample proportions $\hat{p}$ is equal to the population proportion p.

$$\mu_\pi = p = 52\% = 0.52$$

The standard deviation of the sampling distribution of the sample proportion $\hat{p}$ is the square root of the product of p and $(1-p)$ divided by the sample size n.

$$\sigma_\pi = \sqrt{\frac{p(1-p)}{n}} = \sqrt{\frac{0.52(1-0.52)}{3000}} \approx 0.0091$$

The Democratic candidate is leading when the proportion favoring the Republican candidate is actually less than $50\% = 0.05$.

The z-score is the value decreased by the mean, divided by the standard deviation.

$$z = \frac{x-\mu}{\sigma} = \frac{0.50-0.52}{0.0091} \approx -2.20$$

Determine the corresponding probability using the normal probability table in the appendix. $P(0 < Z < 2.20)$ is given in the row starting with 2.2 and in the column starting with .00 of the standard normal probability table in the appendix.

$$\begin{aligned} P(\hat{p} < 0.50) &= P(Z < -2.20) \\ &= P(Z > 2.20) \\ &= 1 - P(Z < 2.20) \\ &= 1 - P(Z < 0) - P(0 < Z < 2.20) \\ &= 1 - 0.5000 - 0.4861 \\ &= 0.0139 \\ &= 1.39\% \end{aligned}$$

Result

(a) $(0.4890, 0.5510)$

(b) $0.1020 = 10.20\%$

(c) $(0.5021, 0.5379)$, $0.0139 = 1.39\%$

Chapter 9 Exercise 67

Page 327

PROBLEM STATEMENT

Chapter 9, Page 327

Solution A

Step 1

A sample of $n = 352$ subscribersshows the mean time spent using the Internet is $\bar{X} = 13.4$ hours per week with a standard deviation of $s = 6.8$ hours.

95 percent Confidence Interval

For a 95 percent confidence interval and degrees of freedom $df = n - 1 = 325 - 1 = 324$, t value is 1.96.

$$\begin{aligned} \bar{X} \pm t\frac{s}{\sqrt{n}} &= 13.4 \pm 1.96 \times \frac{6.8}{\sqrt{352}} \\ &= 13.4 \pm 1.96 \times 0.2025 \\ &= 13.4 \pm 0.3969 \end{aligned}$$

So, $X_1 = 13.4 - 0.3969 = 13.0031$ and $X_2 = 13 + 0.3969 = 13.3969$. So, 95 percent confidence interval for population mean time is $(13.0031, 13.3969)$ hours.

Result

$(13.0031, 13.3969)$ minutes

PROBLEM STATEMENT

Chapter 9, Page 327

Solution A

Step 1

Given:

$$\pi = \text{Population proportion} = 35\% = 0.35$$
$$E = \text{Margin of error} = 2\% = 0.02$$
$$c = \text{Confidence coefficient} = 95\% = 0.95$$

Formula sample size:

$$n = \frac{z^2 \pi (1-\pi)}{E^2}$$

For confidence level $1-\alpha = 0.95$, determine the z-score corresponding to a probability of $1 - 0.5 - \alpha/2 = 1 - 0.5 - 0.05/2 = 0.475$ using table the normal probability table in the appendix:

$$z = 1.96$$

Evaluate the formula for the sample size (round up to the nearest integer!):

$$n = \frac{z^2 \pi (1-\pi)}{E^2} = \frac{1.96^2 \times 0.35(1-0.35)}{0.02^2} \approx 2185$$

Thus we require a sample of size at least 2185.

Result

$$n = 2185$$

Exercise 69

PROBLEM STATEMENT

Chapter 9, Page 327

Solution A

Step 1

Given:

$$n = \text{Sample size} = 105$$
$$c = \text{Confidence level} = 95\% = 0.95$$

(a) **Confidence interval**

The mean is the sum of all values divided by the number of values:

$$\overline{x} = \frac{220.9 + 294.5 + 244.6 + ... + 177.1}{105} = \frac{23215.8}{105} \approx 221.1029$$

The variance is the sum of squared deviations from the mean divided by $n - 1$. The standard deviation is the square root of the variance:

$$s = \sqrt{\frac{(220.9 - 221.1029)^2 + ... + (177.1 - 221.1029)^2}{105 - 1}} \approx 47.1054$$

Determine the t-value by looking in the row starting with degrees of freedom $df = n - 1 = 105 - 1 = 104$ and in the column with $c = 95\%$ in the table of the Student's T distribution (since the table does not contain a row with $df = 104$, we use the nearest smaller degrees of freedom $df = 100$):

$$t_{\alpha/2} = 1.984$$

The margin of error is then:

$$\begin{aligned} E &= t_{\alpha/2} \times \frac{s}{\sqrt{n}} \\ &= 1.984 \times \frac{47.1054}{\sqrt{105}} \\ &\approx 9.1205 \end{aligned}$$

The boundaries of the confidence interval then become:

$$\overline{x} - E = 221.1029 - 9.1205 = 211.9824$$

$$\overline{x} + E = 221.1029 + 9.1205 = 230.2234$$

We are 95% confident that the true population mean selling price of the homes is between 211.9824 thousand dollars and 230.2234 thousand dollars.

Step 2

(b) **Confidence interval**

The mean is the sum of all values divided by the number of values:

$$\overline{x} = \frac{12 + 15 + 9 + ... + 16 + 10}{105} = \frac{1536}{105} = 14.6286$$

The variance is the sum of squared deviations from the mean divided by $n - 1$. The standard deviation is the square root of the variance:

$$s = \sqrt{\frac{(12 - 14.6286)^2 + ... + (10 - 14.6286)^2}{105 - 1}} \approx 4.8739$$

Determine the t-value by looking in the row starting with degrees of freedom $df = n - 1 = 105 - 1 = 104$ and in the column with $c = 95\%$ in the table of the Student's T distribution (since the table does not contain a row with $df = 104$, we use the nearest smaller degrees of freedom $df = 100$):

$$t_{\alpha/2} = 1.984$$

The margin of error is then:

$$\begin{aligned} E &= t_{\alpha/2} \times \frac{s}{\sqrt{n}} \\ &= 1.984 \times \frac{4.8739}{\sqrt{105}} \\ &\approx 0.9437 \end{aligned}$$

The boundaries of the confidence interval then become:

$$\overline{x} - E = 14.6286 - 0.9437 = 13.6849$$

$$\overline{x} + E = 14.6286 + 0.9437 = 15.5723$$

We are 95% confident that the true population mean distance the home is from the center of the city is between 13.6849 miles and 15.5729 miles.

Step 3

(c) Given:

$$x = \text{Number of successes} = 71$$

Confidence interval

The sample proportion is the number of successes divided by the sample size:

$$\pi = \frac{x}{n} = \frac{71}{105} \approx 0.6762$$

For confidence level $1 - \alpha = 0.95$, determine the z-score corresponding to a probability of $1 - 0.5 - \alpha/2 = 1 - 0.5 - 0.05/2 = 0.4750$ using table the normal probability table in the appendix:

$$z = 1.96$$

The margin of error is then:

$$\begin{aligned} E &= z \times \sqrt{\frac{\pi(1-\pi)}{n}} \\ &= 1.96 \times \sqrt{\frac{0.6762(1-0.6762)}{105}} \\ &\approx 0.0895 \end{aligned}$$

The boundaries of the confidence interval then become:

$$\pi - E = 0.6762 - 0.0895 = 0.5867$$

$$\pi + E = 0.6762 + 0.0895 = 0.7657$$

We are 95% confident that the true population proportion of homes with an attached garage is between 0.5867 and 0.7657.

Step 4

(d) We need to summarize the findings from part (a) to (c).

Dear Gary Loftus,

After collecting information about the homes sold in Goodyear, Arizona, we have come to the conclusion that the mean selling price for homes is somewhere between $\$221.982.40$ and $\$230,223.40$ (that is, between 221.9824 thousand dollars and 230.2234 thousand dollars).

Moreover, the mean distance from the home to the center of the city is somewhere between 13.6849 miles and 17.5723 miles.

Finally, the percentage of homes with an attached garage appears to be somewhere between 58.67% and 76.57%.

Kind regards,

Sarah Schrijvers

Result

(a) $(211.9824$ thousand dollars, 230.2234 thousand dollars)

(b) $(13.6849$ miles, 15.5723 miles)

(c) $(0.5867, 0.7657)$

(d) Summarize part (a) to (c)

PROBLEM STATEMENT

Chapter 9, Page 327

Solution A

Step 1

Given:

$$n = \text{Sample size} = 30$$
$$c = \text{Confidence level} = 95\% = 0.95$$

(a) **Confidence interval**

The mean is the sum of all values divided by the number of values:

$$\overline{x} = \frac{160 + 212 + 184 + ... + 156}{30} = \frac{5042}{30} \approx 168.0667$$

The variance is the sum of squared deviations from the mean divided by $n - 1$. The standard deviation is the square root of the variance:

$$s = \sqrt{\frac{(160 - 168.0667)^2 + ... + (156 - 168.0667)^2}{30 - 1}} \approx 33.0735$$

Determine the t-value by looking in the row starting with degrees of freedom $df = n - 1 = 30 - 1 = 29$ and in the column with $c = 95\%$ in the table of the Student's T distribution:

$$t_{\alpha/2} = 2.045$$

The margin of error is then:

$$\begin{aligned} E &= t_{\alpha/2} \times \frac{s}{\sqrt{n}} \\ &= 2.045 \times \frac{33.0735}{\sqrt{30}} \\ &\approx 12.3485 \end{aligned}$$

The boundaries of the confidence interval then become:

$$\overline{x} - E = 168.0667 - 12.3485 = 155.7182$$

$$\overline{x} + E = 168.0667 + 12.3485 = 180.4152$$

We are 95% confident that the true population mean number of home runs per team is between 155.7182 home runs and 180.4152 home runs.

Step 2

(b) **Confidence interval**

The mean is the sum of all values divided by the number of values:

$$\overline{x} = \frac{90 + 82 + 113 + ... + 143}{30} = \frac{2856}{30} = 95.2$$

The variance is the sum of squared deviations from the mean divided by $n - 1$. The standard deviation is the square root of the variance:

$$s = \sqrt{\frac{(90 - 95.2)^2 + ... + (143 - 95.2)^2}{30 - 1}} \approx 15.5949$$

Determine the t-value by looking in the row starting with degrees of freedom $df = n - 1 = 30 - 1 = 29$ and in the column with $c = 95\%$ in the table of the Student's T distribution:

$$t_{\alpha/2} = 2.045$$

The margin of error is then:

$$\begin{aligned} E &= t_{\alpha/2} \times \frac{s}{\sqrt{n}} \\ &= 2.045 \times \frac{15.5949}{\sqrt{30}} \\ &\approx 5.8226 \end{aligned}$$

The boundaries of the confidence interval then become:

$$\overline{x} - E = 95.2 - 5.8226 = 89.3774$$

$$\overline{x} + E = 95.2 + 5.8226 = 101.0226$$

We are 95% confident that the true population mean number of errors committed by each team is between 89.3774 errors and 101.0226 errors.

Step 3

(c) **Confidence interval**

The mean is the sum of all values divided by the number of values:

$$\overline{x} = \frac{76 + 126 + 113 + ... + 75 + 73}{30} = \frac{2970}{30} = 99$$

The variance is the sum of squared deviations from the mean divided by $n - 1$. The standard deviation is the square root of the variance:

$$s = \sqrt{\frac{(76 - 99)^2 + ... + (73 - 99)^2}{30 - 1}} \approx 30.7033$$

Determine the t-value by looking in the row starting with degrees of freedom $df = n - 1 = 30 - 1 = 29$ and in the column with $c = 95\%$ in the table of the Student's T distribution:

$$t_{\alpha/2} = 2.045$$

The margin of error is then:

$$\begin{aligned} E &= t_{\alpha/2} \times \frac{s}{\sqrt{n}} \\ &= 2.045 \times \frac{30.7033}{\sqrt{30}} \\ &\approx 11.4635 \end{aligned}$$

The boundaries of the confidence interval then become:

$$\overline{x} - E = 99 - 11.4635 = 87.5370$$

$$\overline{x} + E = 99 + 11.4635 = 110.4630$$

We are 95% confident that the true population mean number of stolen bases for each team is between 87.5370 stolen bases and 110.4630 stolen bases.

Result

(a) $(155.7182, 180.4152)$

(b) $(89.3774, 101.0226)$

(c) $(87.5370, 110.4630)$

PROBLEM STATEMENT

Chapter 9, Page 328

Solution A

Step 1

Given:

$$n = \text{Sample size} = 80$$
$$c = \text{Confidence level} = 95\% = 0.95$$

(a) **Confidence interval**

The mean is the sum of all values divided by the number of values:

$$\overline{x} = \frac{329 + 503 + 505 + ... + 449}{80} = \frac{36023}{80} = 450.2875$$

The variance is the sum of squared deviations from the mean divided by $n - 1$. The standard deviation is the square root of the variance:

$$s = \sqrt{\frac{(329 - 450.2875)^2 + ... + (449 - 450.2875)^2}{80 - 1}} \approx 53.6860$$

Determine the t-value by looking in the row starting with degrees of freedom $df = n - 1 = 80 - 1 = 79$ and in the column with $c = 95\%$ in the table of the Student's T distribution:

$$t_{\alpha/2} = 1.990$$

The margin of error is then:

$$\begin{aligned} E &= t_{\alpha/2} \times \frac{s}{\sqrt{n}} \\ &= 1.990 \times \frac{53.6860}{\sqrt{80}} \\ &\approx 11.9445 \end{aligned}$$

The boundaries of the confidence interval then become:

$$\overline{x} - E = 450.2875 - 11.9445 = 438.3430$$

$$\overline{x} + E = 450.2875 + 11.9445 = 462.2320$$

We are 95% confident that the true population mean bus maintenance is between $\$438.3430$ and $\$462.2320$.

Step 2

(b) **Confidence interval**

The mean is the sum of all values divided by the number of values:

$$\overline{x} = \frac{853 + 883 + 822 + ... + 449}{80} = \frac{62719}{80} = 783.9875$$

The variance is the sum of squared deviations from the mean divided by $n - 1$. The standard deviation is the square root of the variance:

$$s = \sqrt{\frac{(853 - 137.8850)^2 ... + (449 - 137.8850)^2}{80 - 1}} \approx 137.8850$$

Determine the t-value by looking in the row starting with degrees of freedom $df = n - 1 = 80 - 1 = 79$ and in the column with $c = 95\%$ in the table of the Student's T distribution:

$$t_{\alpha/2} = 1.990$$

The margin of error is then:

$$\begin{aligned} E &= t_{\alpha/2} \times \frac{s}{\sqrt{n}} \\ &= 1.990 \times \frac{137.8850}{\sqrt{80}} \\ &\approx 30.6779 \end{aligned}$$

The boundaries of the confidence interval then become:

$$\overline{x} - E = 783.9875 - 30.6779 = 753.3096$$

$$\overline{x} + E = 783.9875 + 30.6779 = 814.6654$$

We are 95% confident that the true population mean bus miles is between 753.3096 miles and 814.6654 miles.

Step 3

(c) We need to summarize the findings from part (a) and part (b).

Dear State transportation official,

After collecting information about the Buena school district, we have come to the conclusion that the mean bus maintenance cost is between $\$438.34$ and $\$462.23$, while the buses drive on average between 753 miles and 815 miles.

Kind regards,

Sarah Schrijvers

Result

(a) $(438.3430, 462.2320)$

(b) $(753.3096, 814.6654)$

(c) Summarize part (a) and part (b)

Statistics

Chapter 9 — Instructor Solutions Manual

October 24, 2026 • Total Exercises: 13

PROBLEM STATEMENT

Chapter 9, Page 331

Solution A

Step 1

Given:

$$\mu = \text{Mean} = 8.6$$
$$\sigma = \text{Standard deviation} = 2.0$$
$$n = \text{Sample size} = 35$$
$$\overline{x} = 8.8$$

Since the population distribution is normal, the sampling distribution of the sample mean $\overline{x}$ is also normal.

The sampling distribution of the sample mean $\overline{x}$ has mean μ and standard deviation $\dfrac{\sigma}{\sqrt{n}}$.

The z-score is the value decreased by the mean, divided by the standard deviation:

$$z = \frac{\overline{x} - \mu_{\overline{x}}}{\sigma_{\overline{x}}} = \frac{\overline{x} - \mu}{\sigma/\sqrt{n}} = \frac{8.6 - 8.8}{2.0/\sqrt{35}} \approx -0.59$$

Determine the corresponding probability using the normal probability table in the appendix.

$P(Z < 0.59)$ is given in the row starting with 0.5 and in the column starting with .09 of the standard normal probability table in the appendix.

$$\begin{aligned} P(\overline{X} \geq 8.8) &= P(Z > -0.59) \\ &= P(Z < 0.59) \\ &= P(Z < 0) + P(0 < Z < 0.59) \\ &= 0.5000 + 0.2224 \\ &= 0.7224 \\ &= 72.24\% \end{aligned}$$

Result

$$0.7224 = 72.24\%$$

PROBLEM STATEMENT

Chapter 9, Page 331

Solution A

Step 1

Given:

$$\mu = \text{Mean} = 1210$$
$$\sigma = \text{Standard deviation} = 325$$
$$n = \text{Sample size} = 25$$
$$x = 1100$$

Since the population distribution is normal, the sampling distribution of the sample mean $\overline{x}$ is also normal.

The sampling distribution of the sample mean $\overline{x}$ has mean μ and standard deviation $\dfrac{\sigma}{\sqrt{n}}$.

The z-score is the value decreased by the mean, divided by the standard deviation:

$$z = \frac{\overline{x} - \mu_{\overline{x}}}{\sigma_{\overline{x}}} = \frac{\overline{x} - \mu}{\sigma/\sqrt{n}} = \frac{1100 - 1210}{325/\sqrt{25}} \approx -1.69$$

Determine the corresponding probability using the normal probability table in the appendix. $P(Z < 1.69)$ is given in the row starting with 1.6 and in the column starting with .09 of the standard normal probability table in the appendix.

$$\begin{aligned} P(\overline{X} \leq 1100) &= P(Z < -1.69) \\ &= P(Z > 1.69) \\ &= 1 - P(Z < 1.69) \\ &= 1 - P(Z < 0) - P(0 < Z < 1.69) \\ &= 1 - 0.5000 - 0.4545 \\ &= \end{aligned}$$

Result

$$0.0455 = 4.55\%$$

PROBLEM STATEMENT

Chapter 9, Page 331

Solution A

Step 1

Given:

$$\begin{aligned}\overline{x} &= \text{Sample mean} = 160\\ s &= \text{Sample standard deviation} = 20\\ n &= \text{Sample size} = 40\\ c &= \text{Confidence level} = 98\% = 0.98\end{aligned}$$

Confidence interval

Determine the t-value by looking in the row starting with degrees of freedom $df = n - 1 = 40 - 1 = 39$ and in the column with $c = 98\%$ in the table of the Student's T distribution:

$$t_{\alpha/2} = 2.426$$

The margin of error is then:

$$\begin{aligned}E &= t_{\alpha/2} \times \frac{s}{\sqrt{n}}\\ &= 2.426 \times \frac{20}{\sqrt{40}}\\ &\approx 7.6717\end{aligned}$$

The boundaries of the confidence interval then become:

$$\overline{x} - E = 160 - 7.6717 = 152.3283$$

$$\overline{x} + E = 160 + 7.6717 = 167.6717$$

We are 98% confident that the true population mean number of customers per day is between 152.3283 and 167.6717.

Result

$$(152.3283, 167.6717)$$

Chapter 9 **Exercise 4** Page 331

PROBLEM STATEMENT

Chapter 9, Page 331

Solution A

Step 1

Given:

$$\begin{aligned}\overline{x} &= \text{Sample mean} = 2.65\\ s &= \text{Sample standard deviation} = 0.45\\ n &= \text{Sample size} = 20\\ c &= \text{Confidence level} = 90\% = 0.90\end{aligned}$$

Confidence interval

Determine the t-value by looking in the row starting with degrees of freedom $df = n - 1 = 20 - 1 = 19$ and in the column with $c = 90\%$ in the table of the Student's T distribution:

$$t_{\alpha/2} = 1.729$$

The margin of error is then:

$$\begin{aligned}E &= t_{\alpha/2} \times \frac{s}{\sqrt{n}}\\ &= 1.729 \times \frac{0.45}{\sqrt{20}}\\ &\approx 0.1740\end{aligned}$$

The boundaries of the confidence interval then become:

$$\overline{x} - E = 2.65 - 0.1740 = 2.4760$$

$$\overline{x} + E = 2.65 + 0.1740 = 2.8240$$

We are 90% confident that the true population mean waiting time is between 2.4760 minutes and 2.8240 minutes.

Result

$$(2.4760, 2.8240)$$

PROBLEM STATEMENT

Chapter 9, Page 331

Solution A

Step 1

Given:

$$n = \text{Sample size} = 6$$
$$c = \text{Confidence level} = 95\% = 0.95$$

Confidence interval

The mean is the sum of all values divided by the number of values:

$$\overline{x} = \frac{826 + 931 + 1126 + 918 + 1011 + 1101}{6} = \frac{5913}{6} = 985.5$$

The variance is the sum of squared deviations from the mean divided by $n - 1$. The standard deviation is the square root of the variance:

$$s = \sqrt{\frac{(826 - 985.5)^2 + (931 - 985.5)^2 + (1126 - 985.5)^2}{(981 - 985.5)^2 + (1011 - 985.5)^2 + (1101 - 985.5)^2}{6 - 1}} \approx 115.4968$$

Determine the t-value by looking in the row starting with degrees of freedom $df = n - 1 = 6 - 1 = 5$ and in the column with $c = 95\%$ in the table of the Student's T distribution:

$$t_{\alpha/2} = 2.571$$

The margin of error is then:

$$\begin{aligned} E &= t_{\alpha/2} \times \frac{s}{\sqrt{n}} \\ &= 2.571 \times \frac{115.4968}{\sqrt{6}} \\ &\approx 121.2262 \end{aligned}$$

The boundaries of the confidence interval then become:

$$\overline{x} - E = 985.5 - 121.2262 = 864.2738$$

$$\overline{x} + E = 985.5 + 121.2262 = 1106.7262$$

We are 95% confident that the true population mean number of copies per machine is between 864.2738 thousand and 1106.7262 thousand.

🎯 Result

$$(864.2738, 1106.7262)$$

PROBLEM STATEMENT

Chapter 9, Page 331

Solution A

Step 1

Given:

$$n = \text{Sample size} = 5$$
$$c = \text{Confidence level} = 95\% = 0.95 \quad \text{(Assumption)}$$

We note that the sample of the last 5 callers is not a random sample from the population and thus it is likely that the sample is not representative for the population. This then implies that the confidence interval won't be appropriate to calculate as it won't reflect the scope of the true population mean.

Next, we also note that the data values were rounded to the nearest half hour, as the parents didn't measure the number of hours of watching TV precisely. The confidence interval is then not appropriate to be calculated, as the data values won't accurately reflect the population.

Step 2

For those who need to calculate the confidence interval anyway or those who decide otherwise, you find the calculation of the confidence interval below:

Confidence interval

The mean is the sum of all values divided by the number of values:

$$\overline{x} = \frac{3.0 + 3.5 + 4.0 + 4.5 + 3.0}{5} = \frac{18.0}{5} = 3.6$$

The variance is the sum of squared deviations from the mean divided by $n - 1$. The standard deviation is the square root of the variance:

$$s = \sqrt{\frac{(3.0 - 3.6)^2 + (3.5 - 3.6)^2 + (4.0 - 3.6)^2 \\ (4.5 - 3.6)^2 + (3.0 - 3.6)^2}{5 - 1}} \approx 0.6519$$

Determine the t-value by looking in the row starting with degrees of freedom $df = n - 1 = 5 - 1 = 4$ and in the column with $c = 95\%$ in the table of the Students T distribution:

$$t_{\alpha/2} = 2.776$$

The margin of error is then:

$$\begin{aligned} E &= t_{\alpha/2} \times \frac{s}{\sqrt{n}} \\ &= 2.776 \times \frac{0.6519}{\sqrt{5}} \\ &\approx 0.8093 \end{aligned}$$

The boundaries of the confidence interval then become:

$$\overline{x} - E = 3.6 - 0.8093 = 2.7907$$

$$\overline{x} + E = 3.6 + 0.8093 = 4.4093$$

We are 95% confident that the true population mean number of hours of TV watched last night is between 2.7907 hours and 4.4093 hours.

Result

Not appropriate

PROBLEM STATEMENT

Chapter 9, Page 331

Solution A

Step 1

Given:

$$\begin{aligned}\overline{x} &= \text{Sample mean} = 240\\ s &= \text{Sample standard deviation} = 35\\ n &= \text{Sample size} = 16\\ c &= \text{Confidence level} = 95\% = 0.95 \qquad \text{(Assumption)}\end{aligned}$$

Confidence interval

Determine the t-value by looking in the row starting with degrees of freedom $df = n - 1 = 16 - 1 = 15$ and in the column with $c = 95\%$ in the table of the Student's T distribution:

$$t_{\alpha/2} = 2.131$$

The margin of error is then:

$$\begin{aligned}E &= t_{\alpha/2} \times \frac{s}{\sqrt{n}}\\ &= 2.131 \times \frac{35}{\sqrt{16}}\\ &= 18.64625\end{aligned}$$

The boundaries of the confidence interval then become:

$$\overline{x} - E = 240 - 18.64625 = 221.35375$$

$$\overline{x} + E = 240 + 18.64625 = 258.64625$$

We are 90% confident that the true population mean number of widgets produced per day is between 221.35375 and 258.64625.

Since 250 is between 221.35375 and 258.64625, it is likely that the population mean is 250 when the mean daily widget production did not increase and thus it is not reasonable to conclude that the mean daily widget production has increased.

Result

Not reasonable

Chapter 9 **Exercise 8** Page 331

PROBLEM STATEMENT

Chapter 9, Page 331

Solution A

Step 1

Given:

$$\sigma = \text{Population standard deviation} = 0.90$$
$$E = \text{Margin of error} = 0.10$$
$$c = \text{Confidence coefficient} = 95\% = 0.95$$

Formula sample size:

$$n = \left(\frac{z\sigma}{E}\right)^2$$

For confidence level $1 - \alpha = 0.95$, determine the z-score corresponding to a probability of $1 - 0.5 - \alpha/2 = 1 - 0.5 - 0.05/2 = 0.4750$ using table the normal probability table in the appendix:

$$z = 1.96$$

Evaluate the formula for the sample size (round up to the nearest integer!):

$$n = \left(\frac{z\sigma}{E}\right)^2 = \left(\frac{1.96 \times 0.90}{0.10}\right)^2 \approx 312$$

Thus we require a sample of size at least 312.

Result

$$n = 312$$

PROBLEM STATEMENT

Chapter 9, Page 331

Solution A

Step 1

Given:

$$\begin{aligned} \text{Range} &= 5.00 \text{ to } 155.00 \\ E &= \text{Margin of error} = 4.00 \\ c &= \text{Confidence coefficient} = 95\% = 0.95 \end{aligned}$$

We know that nearly all data values are within 2 standard deviations of the mean (when considering a normal distribution). Since the range contains all data values, the range is about 4 standard deviations and thus we estimate the standard deviation as about one fourth of the range.

$$\sigma \approx \frac{\text{Range}}{4} = \frac{155.00 - 5.00}{4} = \frac{150.00}{4} = 37.5$$

Formula sample size:

$$n = \left(\frac{z\sigma}{E}\right)^2$$

For confidence level $1 - \alpha = 0.95$, determine the z-score corresponding to a probability of $1 - 0.5 - \alpha/2 = 1 - 0.5 - 0.05/2 = 0.4750$ using table the normal probability table in the appendix:

$$z = 1.96$$

Evaluate the formula for the sample size (round up to the nearest integer!):

$$n = \left(\frac{z\sigma}{E}\right)^2 = \left(\frac{1.96 \times 37.5}{4.00}\right)^2 \approx 338$$

Thus we require a sample of size at least 338.

Result

$$n = 338$$

PROBLEM STATEMENT

Chapter 9, Page 331

Solution A

Step 1

Given:

$$x = \text{Number of successes} = 120$$
$$n = \text{Sample size} = 200$$
$$c = \text{Confidence level} = 95\% = 0.95$$

Confidence interval

The sample proportion is the number of successes divided by the sample size:

$$\pi = \frac{x}{n} = \frac{120}{200} = 0.6$$

For confidence level $1 - \alpha = 0.95$, determine the z-score corresponding to a probability of $1 - 0.5 - \alpha/2 = 1 - 0.5 - 0.05/2 = 0.4975$ using table the normal probability table in the appendix:

$$z = 1.96$$

The margin of error is then:

$$\begin{aligned} E &= z \times \sqrt{\frac{\pi(1-\pi)}{n}} \\ &= 1.96 \times \sqrt{\frac{0.6(1-0.6)}{200}} \\ &\approx 0.0679 \end{aligned}$$

The boundaries of the confidence interval then become:

$$\pi - E = 0.6 - 0.0679 = 0.5321$$

$$\pi + E = 0.6 + 0.0679 = 0.6679$$

We are 95% confident that the true population proportion of residents who believe the tax rate is too high is between 0.5321 and 0.6679.

We note that 0.50 does not lie in the confidence interval, while the confidence interval contains only values above 0.50. This then implies that we expect more than 50% of the taxpayers to feel that the taxes are too high and thus it is reasonable to conclude that the majority of the taxpayers feel that the taxes are too high.

Result

Reasonable

Chapter 9 **Exercise 11** Page 331

PROBLEM STATEMENT

Chapter 9, Page 331

Solution A

Step 1

Given:

$$\pi = \text{Population proportion} = 8\% = 0.08$$
$$E = \text{Margin of error} = 2\% = 0.02$$
$$c = \text{Confidence coefficient} = 98\% = 0.98$$

Formula sample size:

$$n = \frac{z^2 \pi (1-\pi)}{E^2}$$

For confidence level $1-\alpha = 0.98$, determine the z-score corresponding to a probability of $1 - 0.5 - \alpha/2 = 1 - 0.5 - 0.02/2 = 0.4900$ using table the normal probability table in the appendix:

$$z = 2.33$$

Evaluate the formula for the sample size (round up to the nearest integer!):

$$n = \frac{z^2 \pi (1-\pi)}{E^2} = \frac{2.33^2 \times 0.08(1-0.08)}{0.02^2} \approx 999$$

Thus we require a sample of size at least 999.

Result

$$n = 999$$

PROBLEM STATEMENT

Chapter 9, Page 332

Solution A

Step 1

Given:

$$\pi = \text{Sample proportion} = 25\% = 0.25$$
$$n = \text{Sample size} = 500$$
$$c = \text{Confidence level} = 98\% = 0.98$$

Confidence interval

For confidence level $1 - \alpha = 0.98$, determine the z-score corresponding to a probability of $1 - 0.5 - \alpha/2 = 1 - 0.5 - 0.02/2 = 0.4900$ using table the normal probability table in the appendix:

$$z = 2.33$$

The margin of error is then:

$$\begin{aligned} E &= z \times \sqrt{\frac{\pi(1-\pi)}{n}} \\ &= 2.33 \times \sqrt{\frac{0.25(1-0.25)}{500}} \\ &\approx 0.0451 \end{aligned}$$

The boundaries of the confidence interval then become:

$$\pi - E = 0.25 - 0.0451 = 0.2049$$

$$\pi + E = 0.25 + 0.0451 = 0.2951$$

We are 98% confident that the true population proportion of adults who currently smoke is between 0.2049 and 0.2951.

We note that the confidence interval does not contain 0.30 (30%), while the confidence interval contains only values below 0.30. This then implies that we would agree that the proportion is less than 30 percent.

🎯 Result

$$(0.2049, 0.2951)$$

PROBLEM STATEMENT

Chapter 9, Page 332

Solution A

Step 1

Given:

$$\pi = \text{Population proportion} = 40\% = 0.40$$
$$E = \text{Margin of error} = 3\% = 0.03$$
$$c = \text{Confidence coefficient} = 98\% = 0.98$$

Formula sample size:

$$n = \frac{z^2 \pi (1-\pi)}{E^2}$$

For confidence level $1-\alpha = 0.98$, determine the z-score corresponding to a probability of $1 - 0.5 - \alpha/2 = 1 - 0.5 - 0.02/2 = 0.4900$ using table the normal probability table in the appendix:

$$z = 2.33$$

Evaluate the formula for the sample size (round up to the nearest integer!):

$$n = \frac{z^2 \pi (1-\pi)}{E^2} = \frac{2.33^2 \times 0.40(1-0.40)}{0.03^2} \approx 1448$$

Thus we require a sample of size at least 1448.

Result

$$n = 1448$$

Statistics

Chapter 9 — Instructor Solutions Manual

October 24, 2026 • Total Exercises: 10

Chapter 9 **Exercise 1** Page 332

PROBLEM STATEMENT

Chapter 9, Page 332

Solution A

Step 1

A **random sample** is a sample in which each item of the population had the same chance of being selected.

Result

Random sample

Chapter 9 Exercise 2

Page 332

PROBLEM STATEMENT

Chapter 9, Page 332

Solution A

Step 1

The **sampling error** is the error made when estimating a population parameter by a sample statistic. More precisely, the error is the difference between the population parameter and the sample statistic.

When considering the mean, then the difference between the population mean and the sample mean is thus called the **sampling error** as well.

Result

Sampling error

Chapter 9 Exercise 3

Page 332

PROBLEM STATEMENT

Chapter 9, Page 332

Solution A

Step 1

The standard deviation of the sampling distribution of the sample mean is called the **standard error**.

Result

Standard error

Chapter 9 Exercise 4

Page 332

PROBLEM STATEMENT

Chapter 9, Page 332

Solution A

Step 1

As the sample size increases, we have more information about the population and thus any estimates made about the population will be more accurate.

However, the variance of the sample means measures the accuracy of the sample means in estimating the population mean and thus the variance will **decrease** (as the estimates become more accurate).

Result

Become smaller

Chapter 9 Exercise 5

Page 332

PROBLEM STATEMENT

Chapter 9, Page 332

Solution A

Step 1

A **point estimate** is a single value that is used to estimate some population parameter.

For example, the sample mean is the point estimate of the population mean.

Result

Point estimate

Chapter 9 **Exercise 6** Page 332

PROBLEM STATEMENT

Chapter 9, Page 332

Solution A

Step 1

A **confidence interval** is a range of values in which we expect the true value of the population parameter to lie.

Result

Confidence interval

Chapter 9 **Exercise 7** Page 332

PROBLEM STATEMENT

Chapter 9, Page 332

Solution A

Step 1

The width of the confidence interval is twice the margin of error.

The formula of the margin of error depends on the sample size, the variation in the population, and the level of confidence, while the formula of the margin of error does not depend on the population size (when not using the finite population correction).

This then implies that the width of the confidence interval is not affected by the **size of population**.

Result

Size of population

Chapter 9 **Exercise 8** Page 332

PROBLEM STATEMENT

Chapter 9, Page 332

Solution A

Step 1

A **proportion** is a fraction of a population that has some particular property.

Result

Proportion

Chapter 9 **Exercise 9** Page 332

PROBLEM STATEMENT

Chapter 9, Page 332

Solution A

Step 1

A t distribution is a continuous probability distribution, which has a mean of 0 (that is, the distribution is always centered at 0), and is dependent on the degree of freedom.

However, the t-distribution is not **positively skewed**, because the t distribution is always symmetric (like the normal distribution).

Result

Positively skewed

PROBLEM STATEMENT

Chapter 9, Page 332

Solution A

Step 1

When no preliminary estimate for the population proportion is available, then we use the population proportion 0.50 to determine the sample size because this population proportion will lead to the highest possible sample size (which is thus appropriate for all population proportions).

Result

0.50

Statistics

Chapter 9 — Instructor Solutions Manual

October 27, 2026 • Total Exercises: 4

PROBLEM STATEMENT

Chapter 9, Page 332

Solution A

Step 1

Given:

$$\mu = \text{Mean} = 12.2$$
$$\sigma = \text{Standard deviation} = 2.3$$
$$n = \text{Sample size} = 12$$
$$\overline{x} = 11$$

Since the population distribution is normal, the sampling distribution of the sample mean $\overline{x}$ is also normal.

The sampling distribution of the sample mean $\overline{x}$ has mean μ and standard deviation $\dfrac{\sigma}{\sqrt{n}}$.

The z-score is the value decreased by the mean, divided by the standard deviation:

$$z = \frac{\overline{x} - \mu_{\overline{x}}}{\sigma_{\overline{x}}} = \frac{\overline{x} - \mu}{\sigma/\sqrt{n}} = \frac{11 - 12.2}{2.3/\sqrt{12}} \approx -1.81$$

Determine the corresponding probability using the normal probability table in the appendix. $P(0 < Z < 1.81)$ is given in the row starting with 1.8 and in the column starting with .01 of the standard normal probability table in the appendix.

$$\begin{aligned} P(\overline{X} \leq 11) &= P(Z < -1.81) \\ &= P(Z > 1.81) \\ &= 1 - P(Z < 1.81) \\ &= 1 - P(Z < 0) - P(0 < Z < 1.81) \\ &= 1 - 0.5000 - 0.4649 \\ &= 0.0351 \\ &= 3.51\% \end{aligned}$$

Result

$$0.0351 = 3.51\%$$

PROBLEM STATEMENT

Chapter 9, Page 332

Solution A

Step 1

Given:

$$\begin{aligned}\overline{x} &= \text{Sample mean} = 9.3\\ s &= \text{Sample standard deviation} = 2\\ n &= \text{Sample size} = 26\\ c &= \text{Confidence level} = 90\% = 0.90\end{aligned}$$

(a) The sample mean of 9.3 years refers to the number of years that 26 residents of Conway, SC lived at their current address.

The corresponding population mean then represents the mean number of years that all residents of Conway, SC lived at their current address.

Step 2

(b) The best estimate of the population mean is the sample mean, which is 9.3 years in this case.

Step 3

(c) The standard error of the estimate is the sample standard deviation divided by the square root of the sample size.

$$SE = \frac{s}{\sqrt{n}} = \frac{2}{\sqrt{26}} \approx 0.3922$$

Thus the standard error of the estimate is 0.3922 years.

Step 4

(d) **Confidence interval**

Determine the t-value by looking in the row starting with degrees of freedom $df = n - 1 = 26 - 1 = 25$ and in the column with $c = 90\%$ in the table of the Student's T distribution:

$$t_{\alpha/2} = 1.708$$

The margin of error is then:

$$\begin{aligned} E &= t_{\alpha/2} \times \frac{s}{\sqrt{n}} \\ &= 1.708 \times \frac{2}{\sqrt{26}} \\ &\approx 0.6699 \end{aligned}$$

The boundaries of the confidence interval then become:

$$\overline{x} - E = 9.3 - 0.6699 = 8.6301$$

$$\overline{x} + E = 9.3 + 0.6699 = 9.9699$$

We are 90% confident that the true population mean is between 8.6301 years and 9.699 years.

Result

(a) The mean number of years that all residents of Conway, SC lived at their current address.

(b) 9.3 years

(c) 0.3922 years

(d) $(8.6301, 9.9699)$

PROBLEM STATEMENT

Chapter 9, Page 332

Solution A

Step 1

Given:

$$\pi = \text{Population proportion} = 27\% = 0.27$$
$$E = \text{Margin of error} = 2\% = 0.02$$
$$c = \text{Confidence coefficient} = 98\% = 0.98$$

Formula sample size:

$$n = \frac{z^2 \pi (1-\pi)}{E^2}$$

For confidence level $1-\alpha = 0.98$, determine the z-score corresponding to a probability of $1 - 0.5 - \alpha/2 = 1 - 0.5 - 0.02/2 = 0.4900$ using table the normal probability table in the appendix:

$$z = 2.33$$

Evaluate the formula for the sample size (round up to the nearest integer!):

$$n = \frac{z^2 \pi (1-\pi)}{E^2} = \frac{2.33^2 \times 0.27(1-0.27)}{0.02^2} \approx 2676$$

Thus we require a sample of size at least 2676.

Note: The solution in the back of the book appears to have rounded down to $n = 2675$ instead of rou

Result

$$n = 2676$$

PROBLEM STATEMENT

Chapter 9, Page 332

Solution A

Step 1

Given:

$$x = \text{Number of successes} = 64$$
$$n = \text{Sample size} = 100$$
$$c = \text{Confidence level} = 95\% = 0.95$$

Confidence interval

The sample proportion is the number of successes divided by the sample size:

$$\pi = \frac{x}{n} = \frac{64}{100} = 0.64$$

For confidence level $1 - \alpha = 0.95$, determine the z-score corresponding to a probability of $1 - 0.5 - \alpha/2 = 1 - 0.5 - 0.05/2 = 0.4975$ using table the normal probability table in the appendix:

$$z = 1.96$$

The margin of error is then:

$$\begin{aligned} E &= z \times \sqrt{\frac{\pi(1-\pi)}{n}} \\ &= 1.96 \times \sqrt{\frac{0.64(1-0.64)}{100}} \\ &\approx 0.09408 \end{aligned}$$

The boundaries of the confidence interval then become:

$$\pi - E = 0.64 - 0.09408 = 0.54592$$

$$\pi + E = 0.64 + 0.09408 = 0.73408$$

We are 95% confident that the true population proportion of workers that used public transportation is between 0.54592 and 0.73408.

🎯 Result

$$(0.54592, 0.73408)$$